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GENERAL SPECIFICATIONS

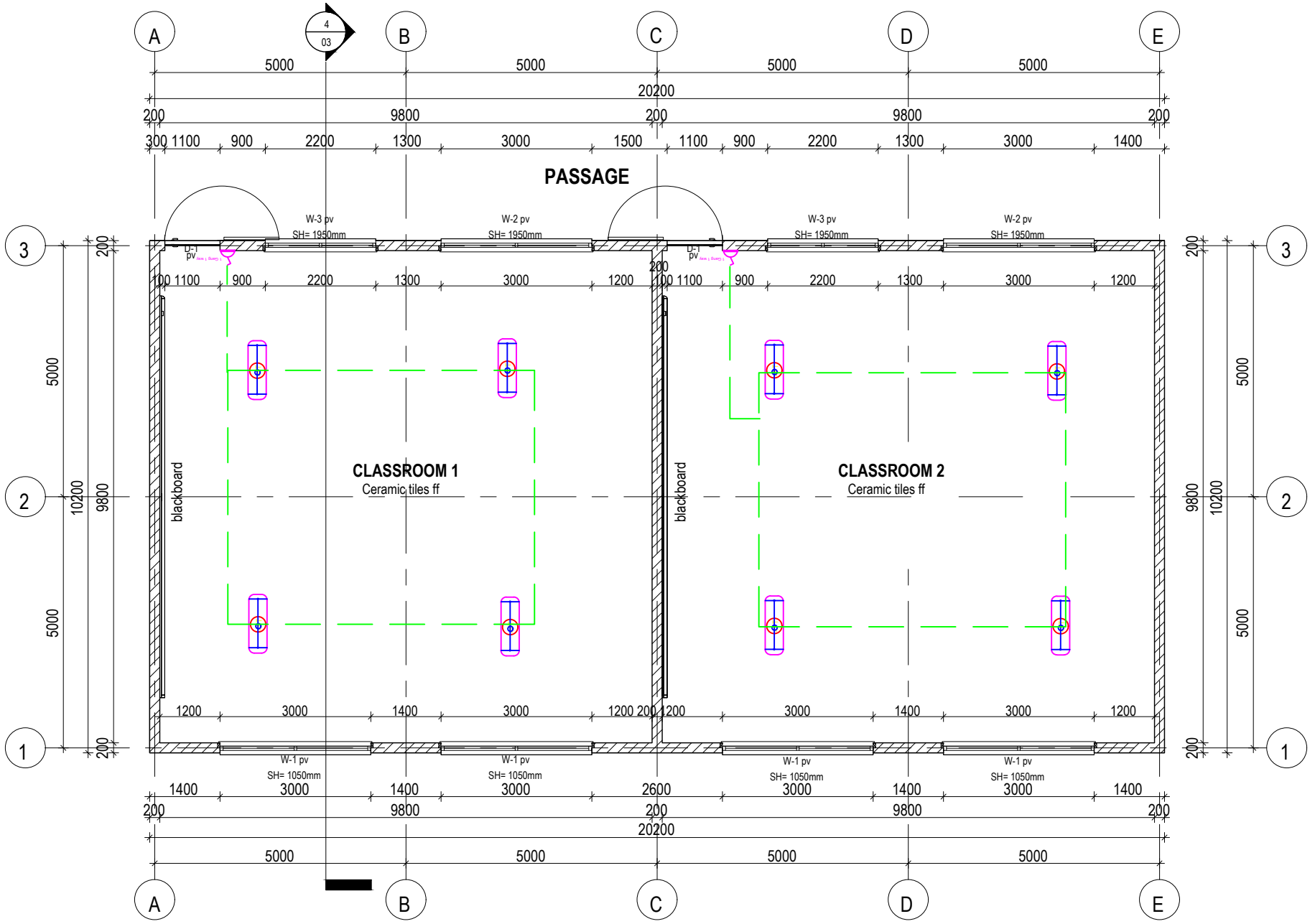
GENERAL NOTES

1. All dimensions are shown in mm unless otherwise specified
2. Drawings are not to be scaled. Only figured dimensions to be used
3. Contractor must check and verify all dimensions on site before commencing of work
4. All slabs at ground level to be poured over 1000gauge polyethene sheet DPM on
5. 50 mm thick some dust on approved grade hardcore filling
6. All soil under the slabs and all around external foundation to be treated with anti-termites insecticide

ELECTRICAL INSTALLATION

1. All electrical installation works are to comply with the specified requirements and should be carried out by a qualified electrician providing requisite test certificates upon completion and installation.
2. Works should be carried out in accordance to approved code of practise

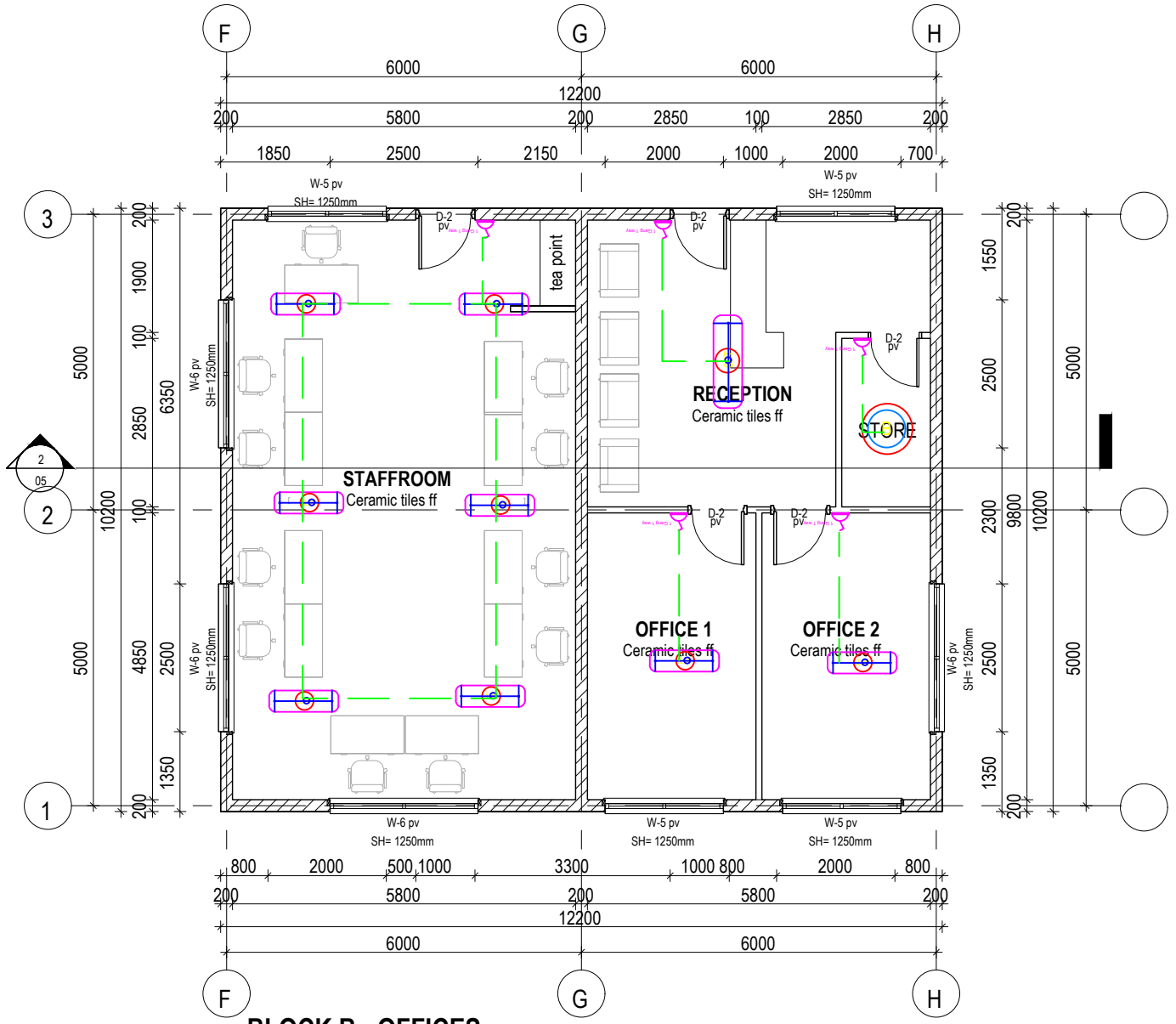
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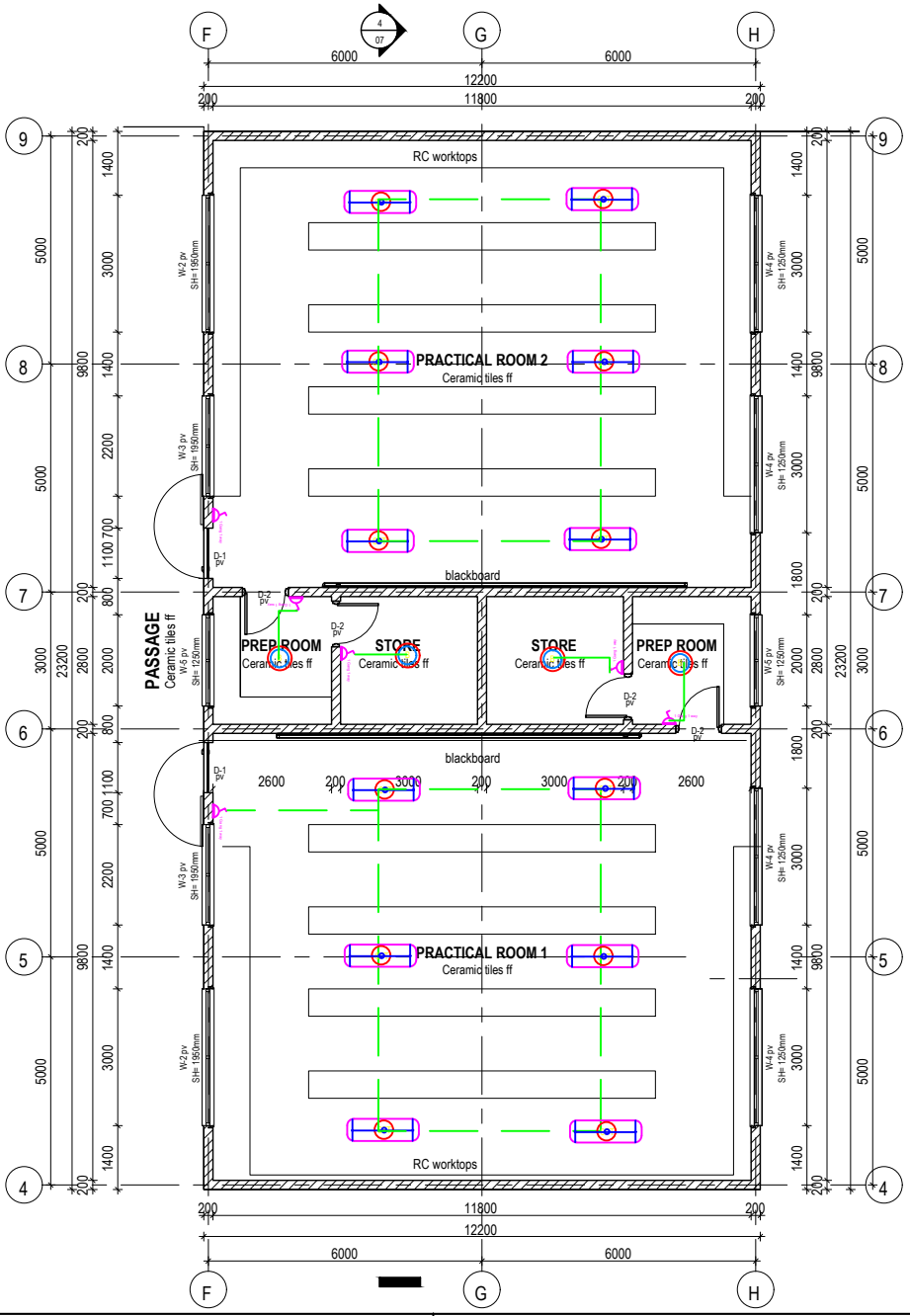
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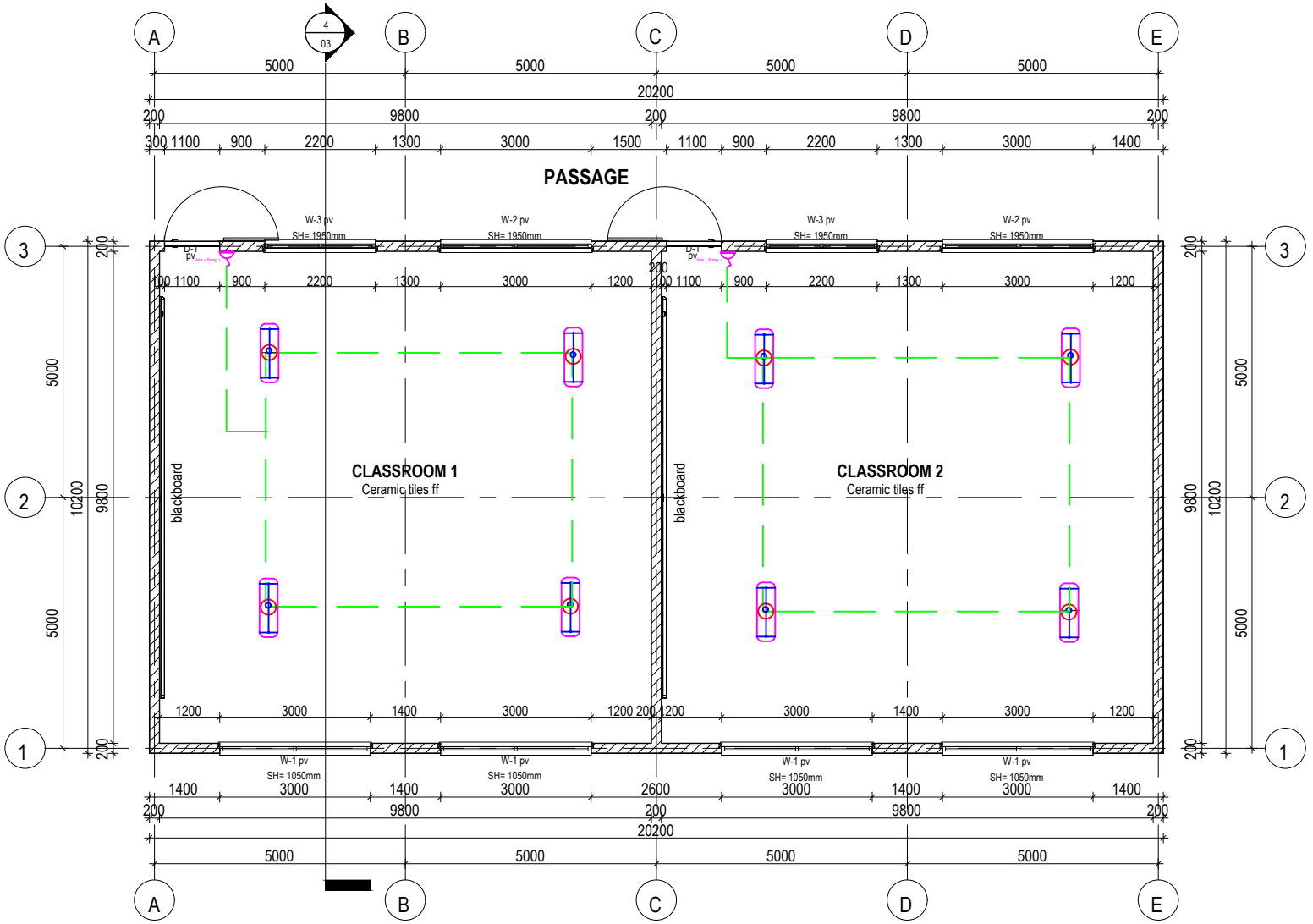
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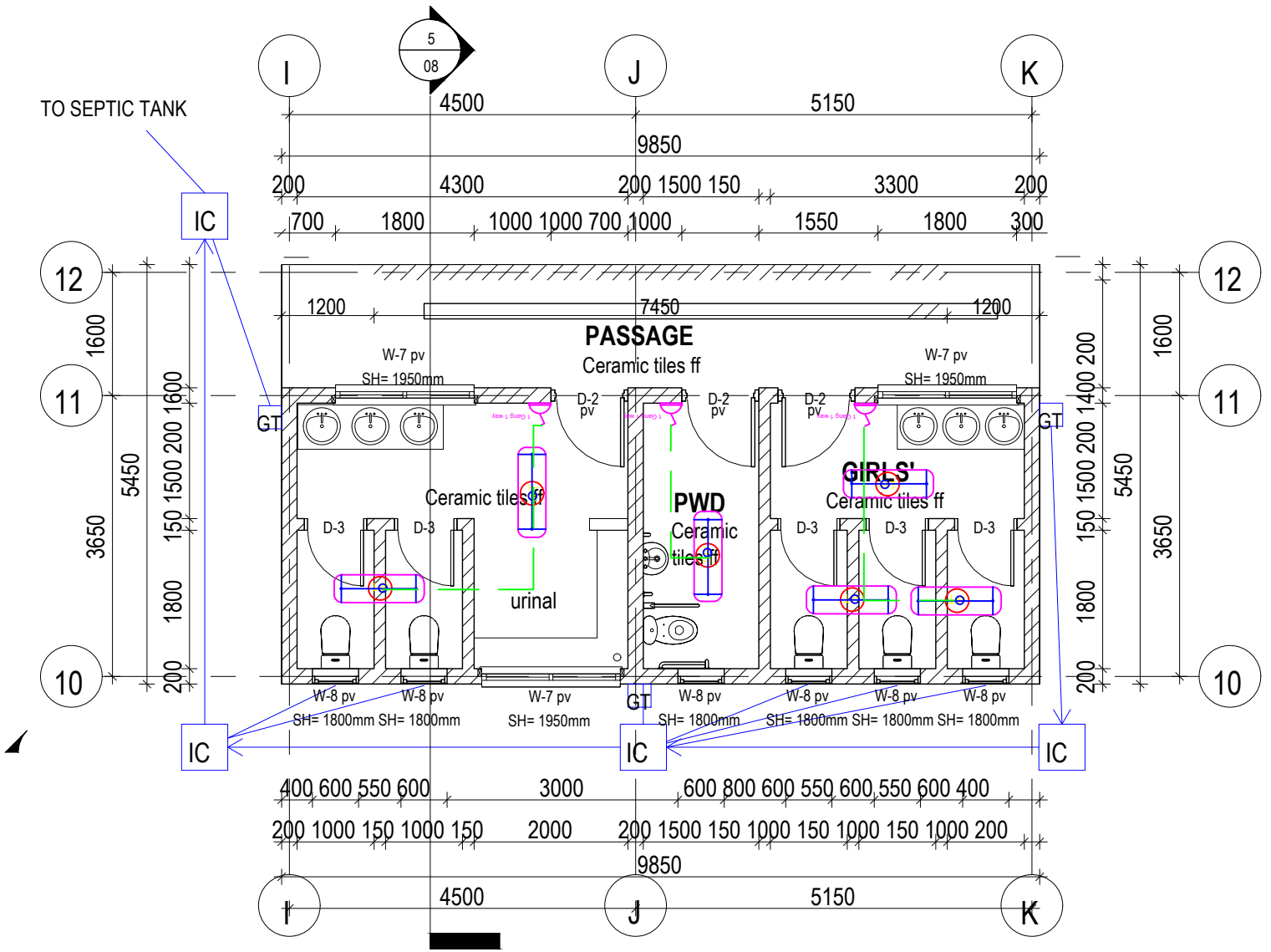
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BLOCK D - CLASSROOMS

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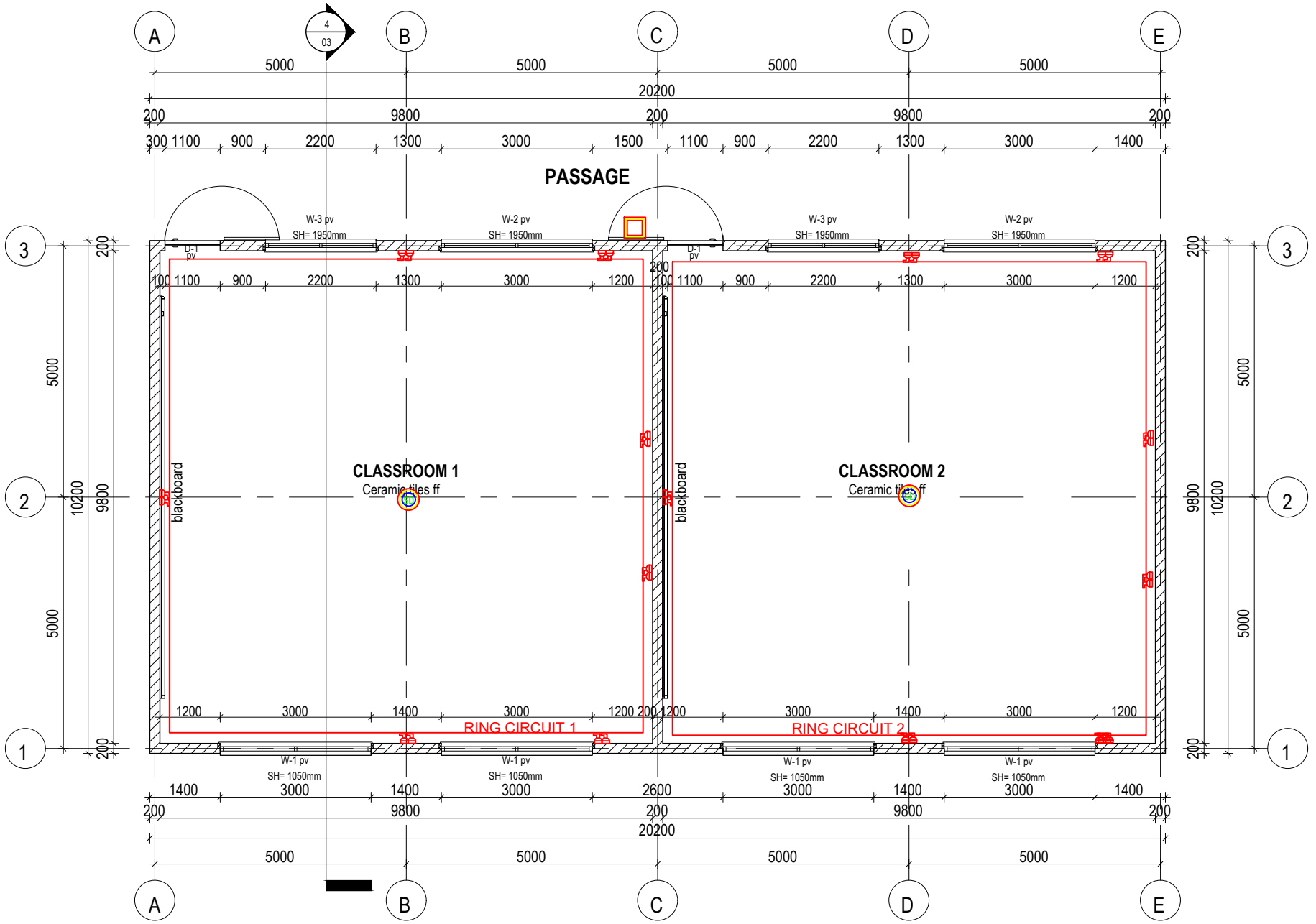


BLOCK E - ABLUTION BLOCK

FLOOR PLAN

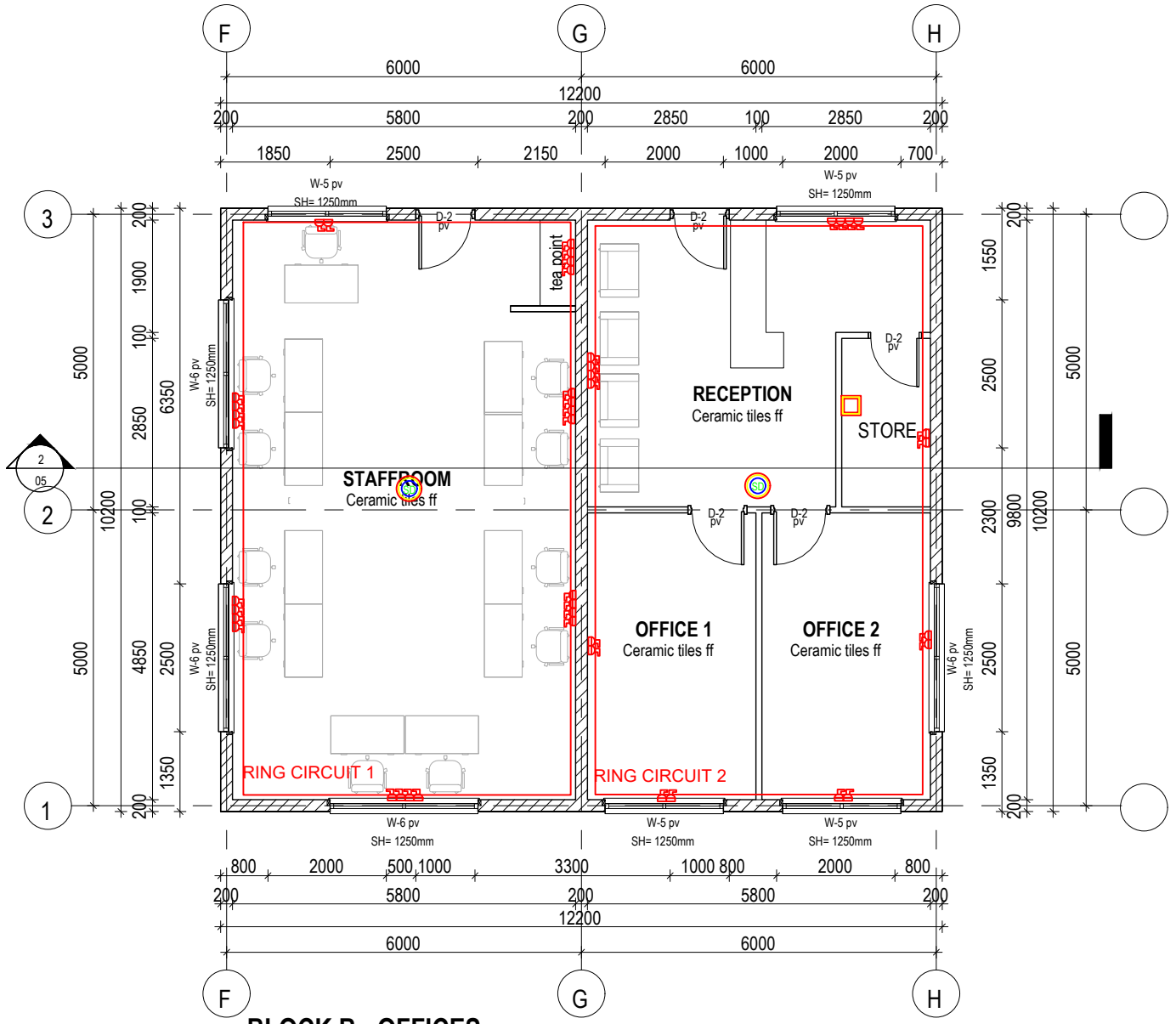
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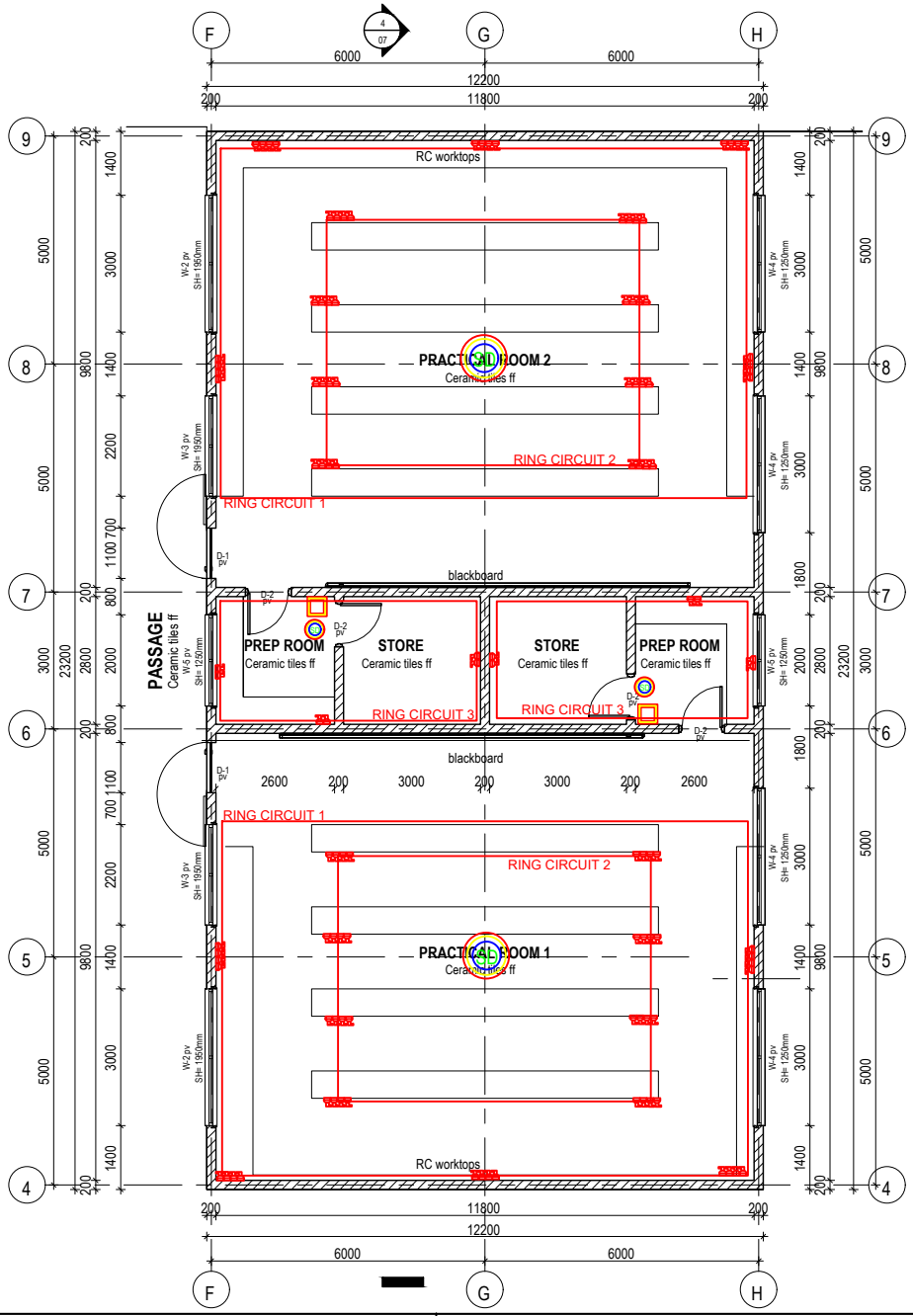
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BLOCK B - OFFICES

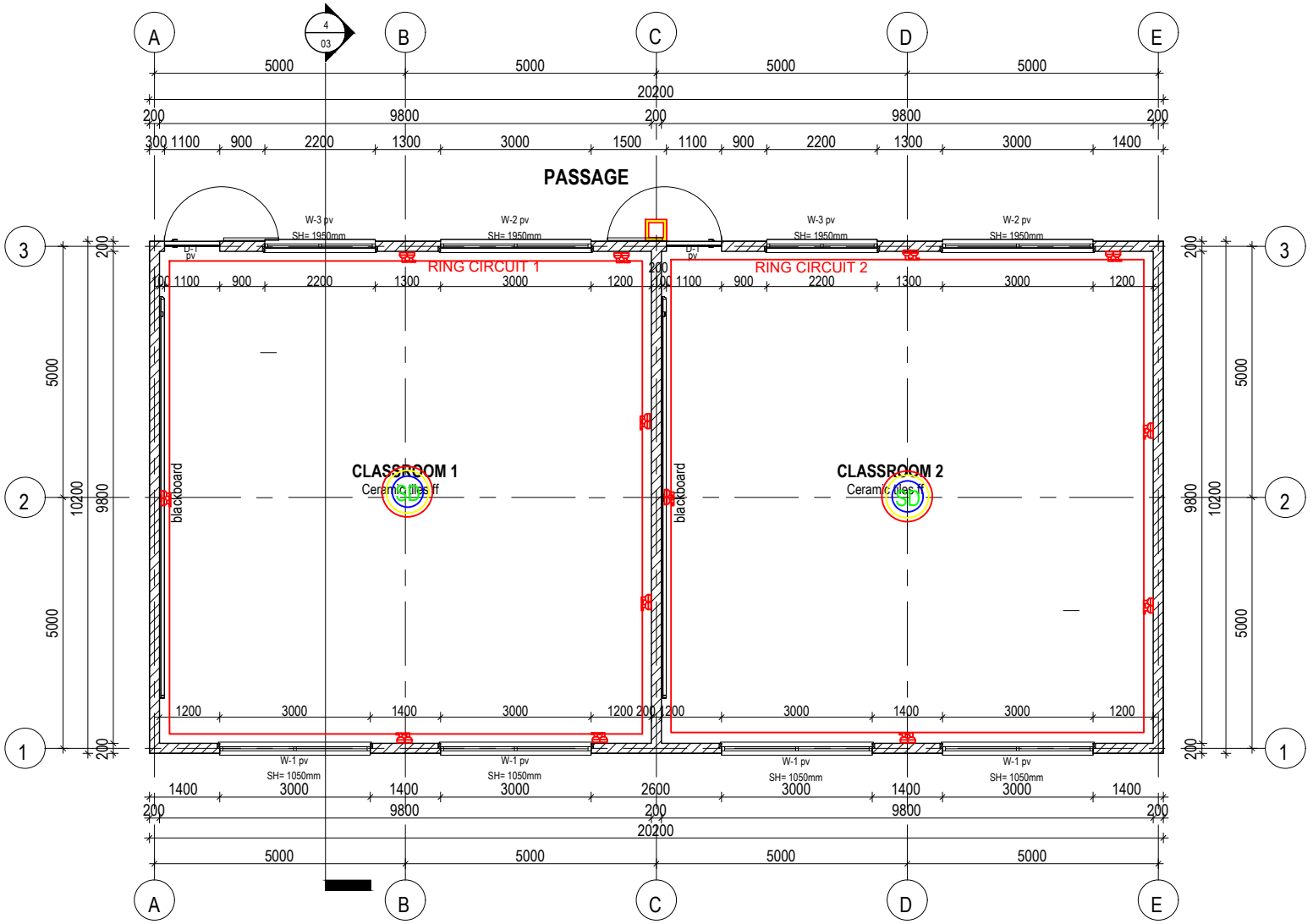
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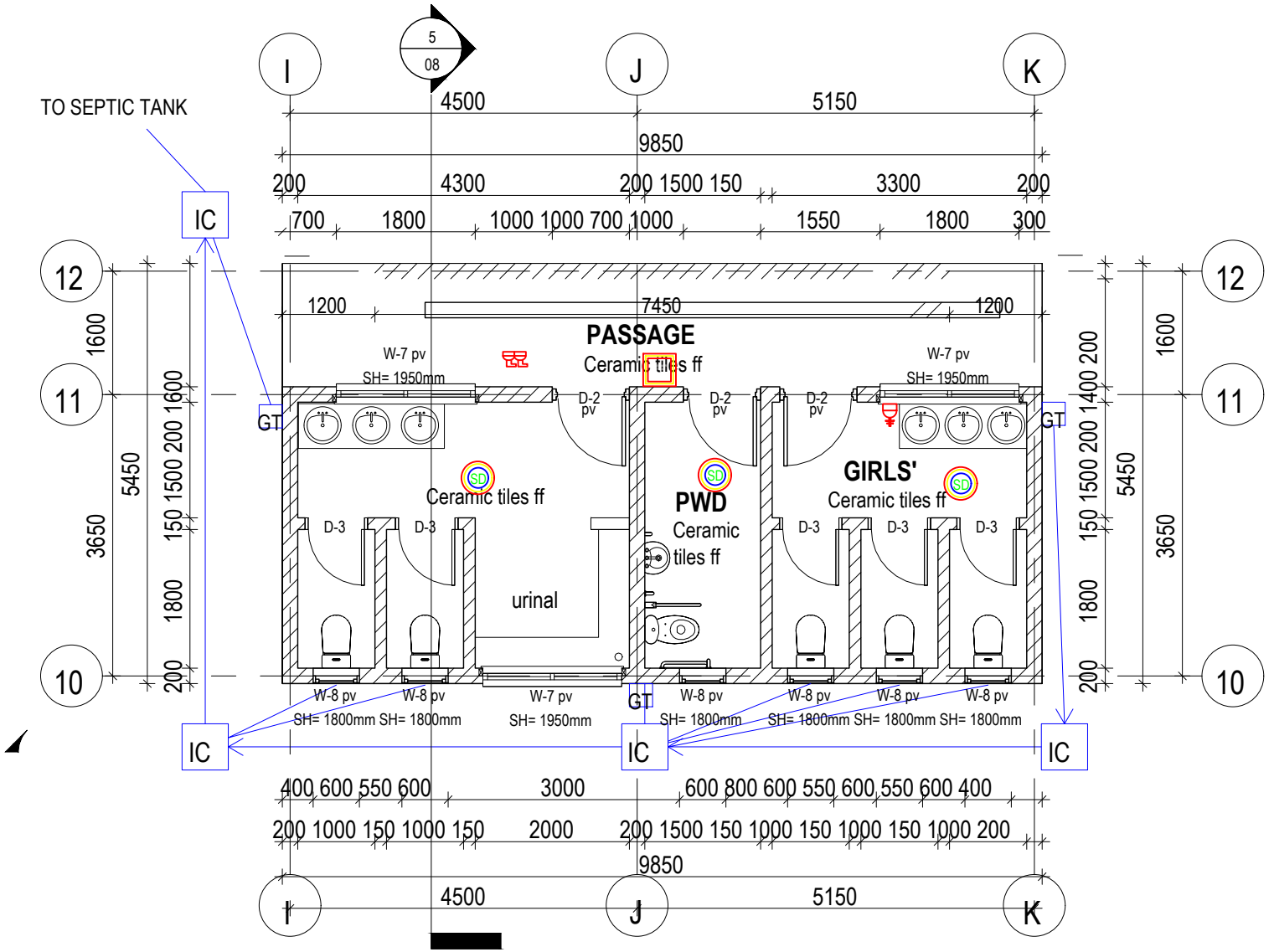
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BLOCK D - CLASSROOMS

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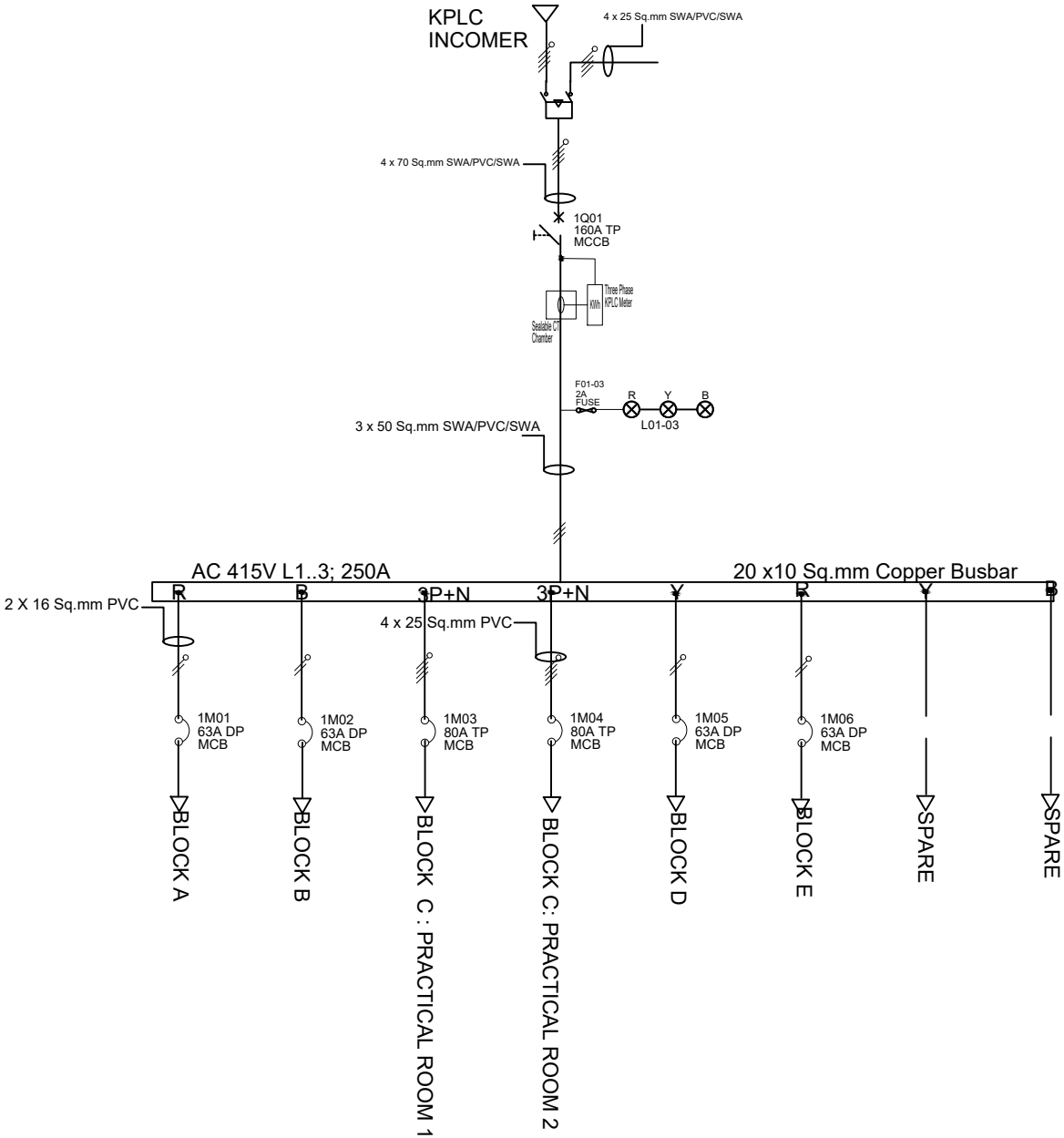


BLOCK E - ABLUTION BLOCK

FLOOR PLAN

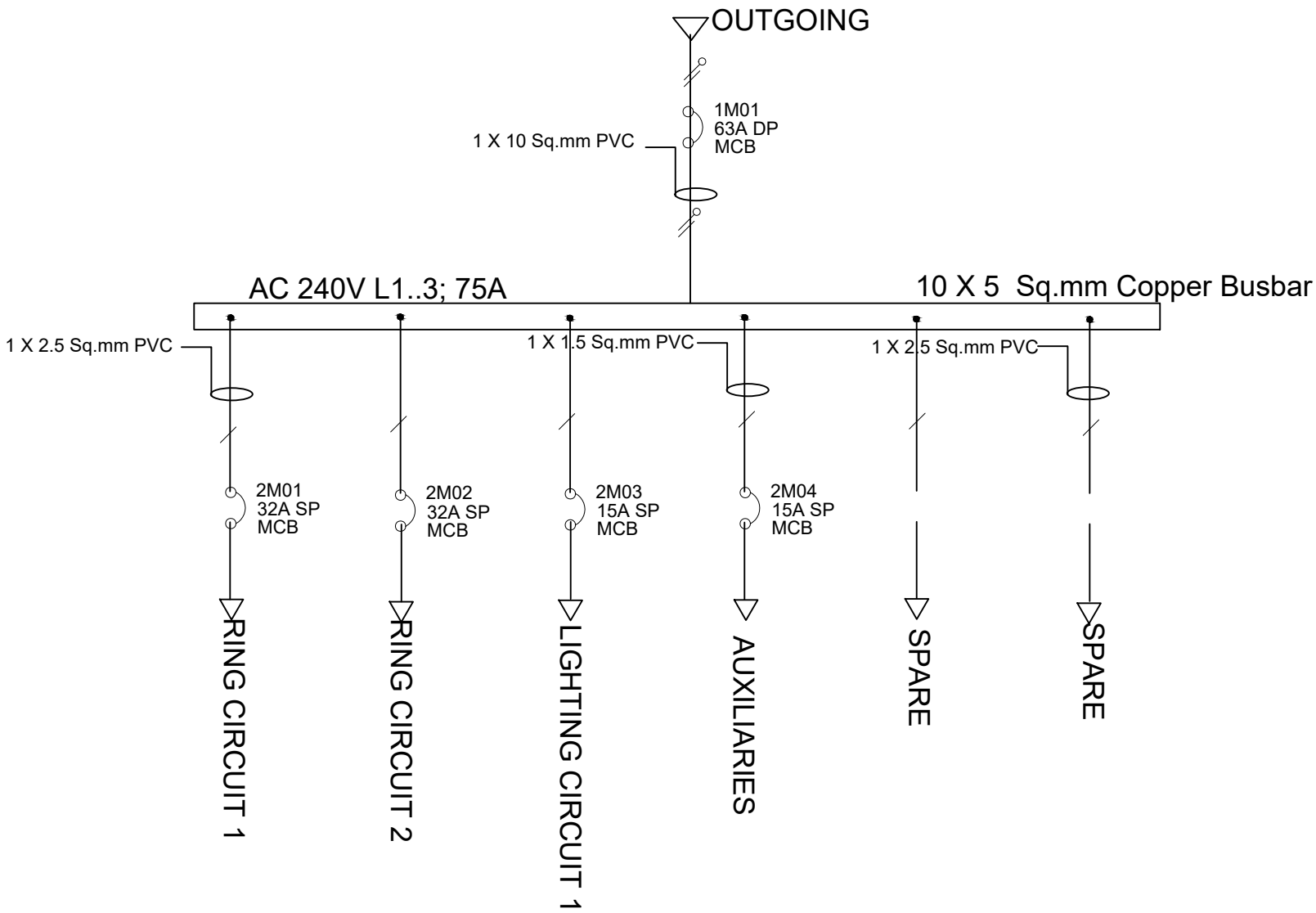
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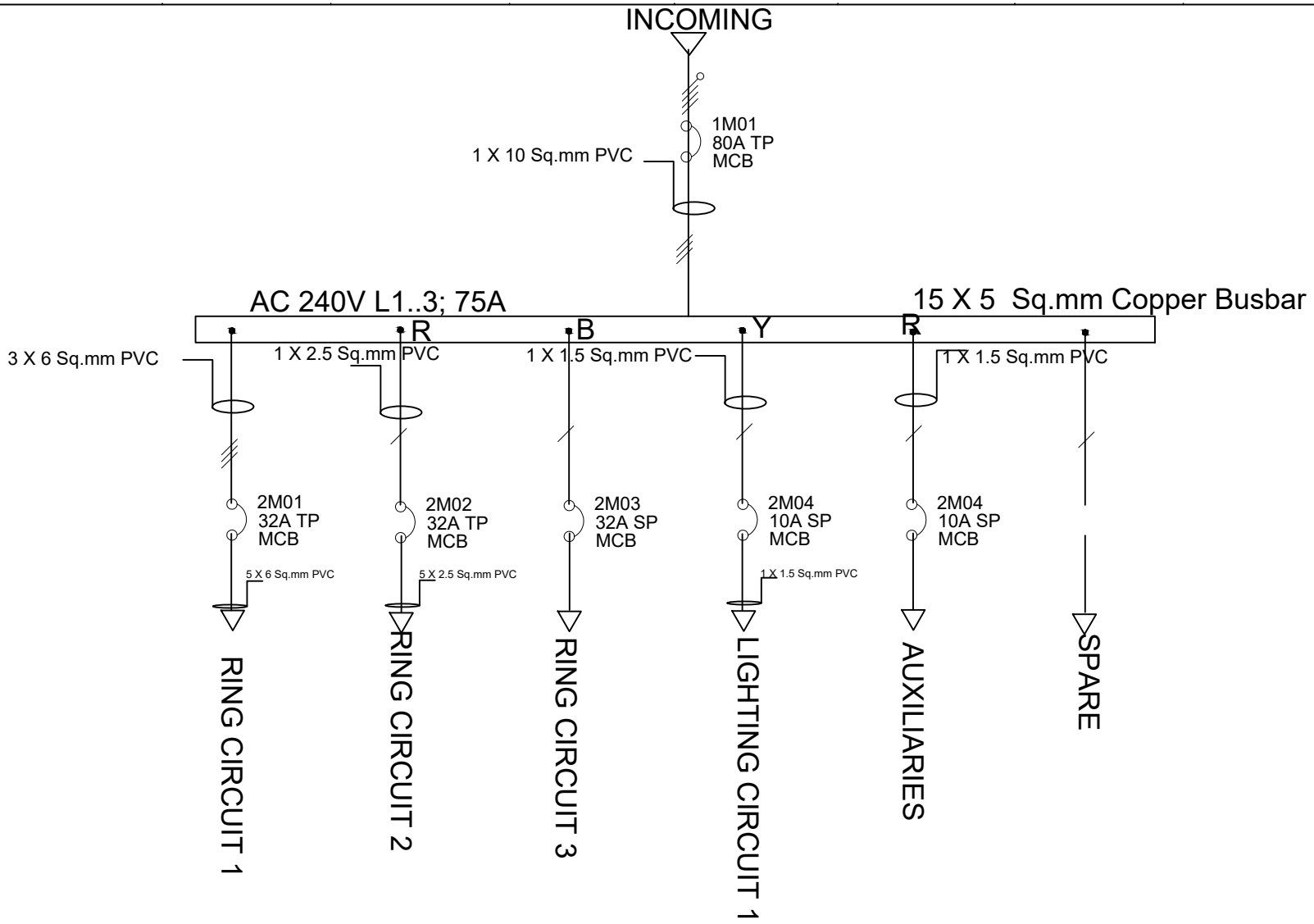
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R1	25/2/2026	SIGNATURE :		PROJECT: HADU VOCATIONAL TRAINIG CENTRE	DATE: 25/2/2026	Sh.No: 1
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R1	25/2/2026		SIGNATURE :	PROJECT: HADU VOCATIONAL TRAINIG CENTRE	DATE: 25/2/2026	Sh.No: 1
R0	19/2/2026		DATE :	CLIENT: ARCHSAN ASSOCIATES	QUOTE REF:Q001	Drg.No: 1

HADU VOCATIONAL TRAINING CENTRE					
	BILL No 1.1 – LIGHTING INSTALLATIONS – GROUND FLOOR				
ITEM	**DESCRIPTION**	**UNIT**	**BQ QTY**	**UnitCost (KSh)**	**TotalCost (Ksh)**
	Indicated Product Brands are only a guide to the quality required				
	Supply, install, test, commission and hand over in proper working condition the following.				
1.11	**LIGHTING POINTS**				
1	Lighting point, wired in 3 x 1.5mm² single core PVC insulated copper cables, drawn in 20mmØ Heavy Duty PVC Conduits concealed in the Building Fabric and complete with bends, couplers and all accessories but excluding the light fitting and switch	No.	8	2,000	16,000
1.12	**LIGHTING ACCESSORIES**				
1	10 Amps single pole molded plate switch for flush mounting as CRABTREE or equal and approved.				
i)	One gang One-way As CRABTREE	No	2	200	400
1.13	**LIGHTING FITTINGS**				
1	**Fitting Slim with LED T8 4ft Tube 50W Day Light** **SKU** FI PR01-50-DL or approved alternative	No	8	750	6,000
	TOTAL FOR BILL No 1.1 - LIGHTING INSTALLATIONS – BLOCK A CARRIED FORWARD TO BILL No 1 COLLECTION PAGE				**22,400.00**
	BILL No 1.2 - LIGHTING				

	INSTALLATIONS - BLOCK B				
ITEM	**DESCRIPTION**	**UNIT**	**BQ QTY**	**Unit Cost (KSh)**	**Total Cost (Ksh)**
1.21	**LIGHTING POINTS**				
1	Lighting point, wired in 3 x 1.5mm² single core PVC insulated copper cables, drawn in 20mmØ Heavy Duty PVC Conduits concealed in the Building Fabric and complete with bends, couplers and all accessories but excluding the light fitting and switch	No	10	2,000	20,000
1.22	**LIGHTING ACCESSORIES**				
1	10 Amps single pole molded plate switch for flush mounting. As Crabtree or equal and approved.				
i)	One gang One-way As CRABTREE	No	8	200	1,600
1.23	**LIGHTING FITTINGS**				
1	**Fitting Slim with LED T8 4ft Tube 50W Day Light** **SKU** FI PR01-50-DL or approved alternative	No	9	750	6,750
	Bulb LED 9 Watts Warm White B22 (Pin) or approved alternative		1	200	200

	TOTAL FOR BILL No 1.2 - LIGHTING INSTALLATIONS - BLOCK B CARRIED FORWARD TO BILL No 1.COLLECTION PAGE				**28,550.00**

BILL No 1.3 - LIGHTING INSTALLATIONS – B L O C K C PRACTICAL ROOM 1

ITEM	**DESCRIPTION**	**UNIT**	**BQ QTY**	**Unit Cost (KSh)**	**Total Cost (Ksh)**
1.31	**LIGHTING POINTS**				

1	Lighting point, wired in 3 x 1.5mm² single core PVC insulated copper cables, drawn in 20mmØ Heavy Duty PVC Conduits concealed in the Building Fabric and complete with bends, couplers and all accessories but excluding the light fitting and switch	No	6	2,000	12,000
1.32	**LIGHTING ACCESSORIES**				
1	10 Amps single pole molded plate switch for flush mounting. As Crabtree or equal and approved.				
i)	One gang One-way As CRABTREE	No	3	200	600

1.33	**LIGHTING FITTINGS**				
1	**Fitting Slim with LED T8 4ft Tube 50W Day Light** **SKU** FI PR01-50-DL or approved alternative	No	7	750	5,250
2	**Bulb LED 9 Watts Warm White B22 (Pin)** or approved alternative	No	1	200	200
	TOTAL FOR BILL No 1.3 - LIGHTING INSTALLATIONS BLOCK C : PRACTICAL ROOM 1 -CARRIED FORWARD TO BILL No COLLECTION PAGE				**18,050**
	BILL No 1.4 - LIGHTING INSTALLATIONS – B L O C K C PRACTICAL ROOM 2				
ITEM	**DESCRIPTION**	**UNIT**	**BQ QTY**	**Unit Cost (KSh)**	**Total Cost (Ksh)**
1.41	**LIGHTING POINTS**				
1	Lighting point, wired in 3 x 1.5mm² single core PVC insulated copper cables, drawn in 20mmØ Heavy Duty PVC Conduits concealed in the Building Fabric and complete with bends, couplers and all accessories but excluding the light fitting and switch	No	6	2,000	12,000
1.42	**LIGHTING ACCESSORIES**				

	10 Amps single pole molded plate switch for flush mounting. As Crabtree or equal and approved.				
i)	One gang One-way As CRABTREE	No	3	200	600
1.43	**LIGHTING FITTINGS**				
1	**Fitting Slim with LED T8 4ft Tube 50W Day Light** **SKU** FI PR01-50-DL or approved alternative	No	7	750	5,250
2	**Bulb LED 9 Watts Warm White B22 (Pin)** or approved alternative	No	1	200	200
	TOTAL FOR BILL No 1.4 - LIGHTING INSTALLATIONS BLOCK C : PRACTICAL ROOM 1 -CARRIED FORWARD TO BILL No 1 COLLECTION PAGE				**18,050.00**
	BILL No 1.5 - LIGHTING INSTALLATIONS – B L O C K D				
ITEM	**DESCRIPTION**	**UNIT**	**BQ QTY**	**Unit Cost (KSh)**	**Total Cost (Ksh)**
1.51	**LIGHTING POINTS**				
1	Lighting point, wired in 3 x 1.5mm² single core PVC insulated copper cables, drawn in 20mmØ Heavy Duty PVC Conduits concealed in the Building Fabric and complete with bends, couplers and all accessories but excluding the light fitting and switch	No	8	2,000	16,000
1.52	**LIGHTING ACCESSORIES**				
	10 Amps single pole molded plate switch for flush mounting. As Crabtree or equal and approved.				
i)	One gang One-way As CRABTREE	No	2	200	400
1.53	**LIGHTING FITTINGS**				
1	**Fitting Slim with LED T8 4ft Tube 50W Day Light** **SKU** FI PR01-50-DL or approved	No	8	750	6,000

	alternative				
	TOTAL FOR BILL No 1.5 - LIGHTING INSTALLATIONS 2nd FLOOR -CARRIED FORWARD TO BILL No 2 COLLECTION PAGE				**22,400.00**
	BILL No 1.6 - LIGHTING INSTALLATIONS – B L O C K E				
ITEM	**DESCRIPTION**	**UNIT**	**BQ QTY**	**Unit Cost (KSh)**	**Total Cost (Ksh)**
1.61	**LIGHTING POINTS**				
1	Lighting point, wired in 3 x 1.5mm² single core PVC insulated copper cables, drawn in 20mmØ Heavy Duty PVC Conduits concealed in the Building Fabric and complete with bends, couplers and all accessories but excluding the light fitting and switch	No	5	2,000	10,000
1.62	**LIGHTING ACCESSORIES**				
	10 Amps single pole molded plate switch for flush mounting. As Crabtree or equal and approved.				
i)	One gang One-way As CRABTREE	No	3	200	400
1.63	**LIGHTING FITTINGS**				
1	**Fitting Slim with LED T8 4ft Tube 50W Day Light** **SKU** FI PR01-50-DL or approved alternative	No	5	750	3,750
	TOTAL FOR BILL No 1.6 - LIGHTING INSTALLATIONS 2nd FLOOR -CARRIED FORWARD TO BILL No 1 COLLECTION PAGE				**14,150.00**
ITEM	**DESCRIPTION**				**BQ AMOUNT**
1	SUB-TOTAL FOR BILL No 1.1 - LIGHTING INSTALLATIONS – BLOCK A				22,400

2	SUB-TOTAL FOR BILL No 1.2 - LIGHTING INSTALLATIONS – BLOCK B	28,550
3	SUB-TOTAL FOR BILL No 1.3- LIGHTING INSTALLATIONS – BLOCK C PRACTICAL ROOMS	18,050
4	SUB-TOTAL FOR BILL No 1.4- LIGHTING INSTALLATIONS – BLOCK C PRACTICAL ROOM 2	18,050
5	SUB-TOTAL FOR BILL No 1.5- LIGHTING INSTALLATIONS – BLOCK D	22,400
6	SUB-TOTAL FOR BILL No 1.5- LIGHTING INSTALLATIONS – BLOCK D	14,150

TOTAL FOR BILL No 1 - LIGHTING INSTALLATIONS CARRIED FORWARD TO THE PRICES SUMMARY PAGE	**95,050.00**

BILL No 2.1 - POWER INSTALLATIONS – BLOCK A					
ITEM	**DESCRIPTION**	**UNIT**	**BQ QTY**	**Unit Cost (KSh)**	**Total Cost (Ksh)**
	Indicated Product Brands are only a guide to the quality required				
	Supply, install, test, commission and hand over in proper working condition the following.				
2.11	**POWER POINTS**				
1	13 Amps Socket Outlet Point, recessed in the wall wired in 3 x 2.5mm² single core PVC insulated copper cables in ring circuit drawn in 25mmØ Heavy Duty PVC Conduits concealed in the Building Fabric, from the Raw Power Distribution Board but without the outlet plate.	No	14	3,000	42,000
2.12	POWER ACCESSORIES				
1	13 Amps Twin Switched double pole socket outlet plate for flush mounting and connection to raw power supply. As CRABTREE Catalogue No 9704 or equal and approved.	No	14	600	8,400

TOTAL FOR BILL No 2.1 - POWER INSTALLATIONS - GROUND FLOOR CARRIED FORWARD TO BILL No. 2 COLLECTION PAGE	**50,400.00**

	BILL No. 2.2 - POWER INSTALLATIONS – BLOCK B				
ITEM	**DESCRIPTION**	**UNIT**	**BQ QTY**	**Unit Cost (KSh)**	**Total Cost (Ksh)**
2.21	**POWER POINTS**				

1	13 Amps Socket Outlet Point, recessed in the wall wired in 3 x 2.5mm² single core PVC insulated copper cables in ring circuit drawn in 25mmØ Heavy Duty PVC Conduits concealed in the Building Fabric, from the Raw Power Distribution Board but without the outlet plate.	No	22	3,000	66,000
2.22	**POWER ACCESSORIES**				
1	13 Amps Twin Switched double pole socket outlet plate for flush mounting and connection to raw power supply. As CRABTREE Catalogue No 9704 or equal and aproved.	No	22	600	13,200
TOTAL FOR BILL No 2.2 - POWER INSTALLATIONS FOR BLOCK B - CARRIED FORWARD TO BILL No. 2 COLLECTION PAGE					**78,200.00**
BILL No. 2.3 - POWER INSTALLATIONS FOR – BLOCK C : PRACTICAL ROOM 1					
ITEM	**DESCRIPTION**	**UNIT**	**BQ QTY**	**Unit Cost (KSh)**	**Total Cost (Ksh)**
2.31	**POWER POINTS**				
1	13 Amps Socket Outlet Point, recessed in the wall wired in 3 x 2.5mm² single core PVC insulated copper cables in ring circuit drawn in 25mmØ Heavy Duty PVC Conduits concealed in the Building Fabric, from the Raw Power Distribution Board but without the outlet plate.	No	11	3,000	33,000
2	16Amps 3 Phase power circuit Industrial outlet point comprising of 5x6.0mm² single core PVCI copper cables drawn in concealed PVC conduit	No	10	4,000	40,000
2.32	**POWER ACCESSORIES**				
1	13 Amps Twin Switched double pole socket outlet plate for flush mounting and connection to raw power supply. As CRABTREE Catalogue No 9704 or equal and approved.	No	22	600	13,200
2	**Industrial Plug 3 Pin 16 Amps SKU** PS PL3P-16-44	No	20	1,000	20,000

TOTAL FOR BILL No 2.3 - POWER INSTALLATIONS -PRACTICAL ROOM 1 CARRIED FORWARD TO BILL No. 2 COLLECTION PAGE					**106,200**
BILL No. 2.4 - POWER INSTALLATIONS FOR – BLOCK C : PRACTICAL ROOM 2					
ITEM	**DESCRIPTION**	**UNIT**	**BQ QTY**	**Unit Cost (KSh)**	**Total Cost (Ksh)**
2.41	**POWER POINTS**				
1	13 Amps Socket Outlet Point, recessed in the wall wired in 3 x 2.5mm² single core PVC insulated copper cables in ring circuit drawn in 25mmØ Heavy Duty PVC Conduits concealed in the Building Fabric, from the Raw Power Distribution Board but without the outlet plate	No	11	3,000	33,000
2	16Amps 3 Phase power circuit Industrial outlet point comprising of 5x6.0mm² single core PVCI copper cables drawn in concealed PVC conduit	No	10	4,000	40,000
2.42	**POWER ACCESSORIES**				
1	13 Amps Twin Switched double pole socket outlet plate for flush mounting and connection to raw power supply. As CRABTREE Catalogue No 9704 or equal and approved.	No	22	600	13,200
2	**Industrial Plug 3 Pin 16 Amps SKU** PS PL3P-16-44	No	20	1000	20,000
	TOTAL FOR BILL No 2.4 - POWER INSTALLATIONS - PRACTICAL ROOM 1 CARRIED FORWARD TO BILL No. 2 COLLECTION PAGE				**106,200.00**
	BILL No. 2. 5 - POWER INSTALLATIONS FOR – BLOCK D				
ITEM	**DESCRIPTION**	**UNIT**	**BQ QTY**	**Unit Cost (KSh)**	**Total Cost (Ksh)**
2.51	**POWER POINTS**				
1	13 Amps Socket Outlet Point, recessed in the wall wired in 3 x	No	14	3,000	42,000

	2.5mm² single core PVC insulated copper cables in ring circuit drawn in 25mmØ Heavy Duty PVC Conduits concealed in the Building Fabric, from the Raw Power Distribution Board but without the outlet plate				
2.52	**POWER ACCESSORIES**				
1	13 Amps Twin Switched double pole socket outlet plate for flush mounting and connection to raw power supply. As CRABTREE Catalogue No 9704 or equal and approved.	No	14	600	8,400
	TOTAL FOR BILL No 2.5 - POWER INSTALLATIONS - BLOCK D CARRIED FORWARD TO BILL No. 2 COLLECTION PAGE				**50,400.00**
	BILL No. 2. 6 - POWER INSTALLATIONS FOR – BLOCK E				
ITEM	**DESCRIPTION**	**UNIT**	**BQ QTY**	**Unit Cost (KSh)**	**Total Cost (Ksh)**
2.61	**POWER POINTS**				
1	13 Amps Socket Outlet Point, recessed in the wall wired in 3 x 2.5mm² single core PVC insulated copper cables in ring circuit drawn in 25mmØ Heavy Duty PVC Conduits concealed in the Building Fabric, from the Raw Power Distribution Board but without the outlet plate	No	3	3,000	9,000
2.62	**POWER ACCESSORIES**				
1	13 Amps Twin Switched double pole socket outlet plate for flush mounting and connection to raw power supply. As CRABTREE Catalogue No 9704 or equal and approved.	No	3	600	600
	TOTAL FOR BILL No 2.5 - POWER INSTALLATIONS - BLOCK E CARRIED FORWARD TO BILL No. 2 COLLECTION PAGE				**10,800.00**

BILL No. 3 - POWER INSTALLATIONS COLLECTION PAGE					
ITEM	**DESCRIPTION**				**BQ AMOUNT**
1	SUB-TOTAL FOR BILL No 2.1 - POWER POINTS- BLOCK A				50,400.00
2	SUB-TOTAL FOR BILL No 2.2 - POWER POINTS- BLOCK B				78, 200.00
3	SUB-TOTAL FOR BILL No 2.3 - POWER POINTS – BLOCK C :PRACTICAL ROOM 1				106,200.00
4	SUB-TOTAL FOR BILL No 2.4 - POWER POINTS – BLOCK C :PRACTICAL ROOM 2				106,200.00
5	SUB-TOTAL FOR BILL No 2.5 - POWER POINTS – BLOCK D				50,400.00
6	SUB-TOTAL FOR BILL No 2.6 - POWER POINTS – BLOCK E :PRACTICAL ROOM 1				10,800.00
TOTAL FOR BILL No. 2 - POWER INSTALLATIONS CARRIED FORWARD TO THE PRICES SUMMARY PAGE					**402,200.00**

BILL No. 3.1 - ELECTRICAL POWER DISTRIBUTION -					
ITEM	**DESCRIPTION**	**UNIT**	**BQ QTY**	**Unit Cost (KSh)**	**Total Cost (Ksh)**
1	Recessed, TPN MCB Distribution Board incorporating 160 Amps TPN integral isolator, lockable front door, labelling front cover and spare way blanking plates. As HAGER , MERLIN GERIN or equal and approved.	LOT	1		

ITEM	**DESCRIPTION**	**UNIT**	**BQ QTY**	**Unit Cost (KSh)**	**Total Cost (Ksh)**

i)	8-way TPN MCB Distribution Board cw 80A integral isolator	No	1	15,000	15,000
ii)	6-way TPN MCB Distribution Board cw 63A integral isolator	No	6	12,000	72,000
2	Miniature Circuit Breaker for incorporation in Distribution Boards above As HAGER , MERLIN GERIN or equal and approved.				
(i)	10 Amps SP	No	6	500	3,000
(ii)	32 Amps SP	No	12	1,000	12,000
(iii)	32A TP	No	2	2,000	4,000
(iv)	63A DP	No	4	1,500	6,000
(iv)	80A TP	No	2	1,500	3,000
(v)	Single Blanking plates	No	15	200	3,000

Sub-Total for Bill No 3.1 -ELECTRICAL POWER DISTRIBUTION - carried to next Page; Ksh	118,000

ITEM	**DESCRIPTION**	**UNIT**	**BQ QTY**	**Unit Cost (KSh)**	**Total Cost (Ksh)**
Sub-Total for Bill No 3.1 -ELECTRICAL POWER DISTRIBUTION - Brought from Previous Page; Ksh					**118,000.00**

3	Sub-mains comprising XLPE/SWA/PVC cable complete with glands, surface laid with saddles, in a Perfoma Mesh Tray from the LV Switch Board to the Distribution Boards in the Ground Floor as follows:				
i)	16.0mm², SC core XLPE/PVC SC copper cable to Distribution Board ON BLOCK A	LM	20	1,800	36,000
ii)	16.0mm², SC core XLPE/PVC SC copper cable to Distribution Board ON BLOCK B	LM	20	1,800	36,000
iii)	16.0mm², SC core XLPE/PVC SC copper cable to Distribution Board ON BLOCK D	LM	50	1,800	90,000
iv)	25.0mm²,4 core XLPE/SWA/PVC 4 copper cable to Distribution Board ON BLOCK C	LM	10	1,800	18,000
v)	16.0mm², SC core XLPE/PVC SC copper cable to Distribution Board ON BLOCK E	LM	15	1,800	27,000
4	Heavy Duty PVC Conduits concealed in the Building Fabric complete with bends,couplers and all other necessary accessories as follows:-				
i)	16 mmØ for Mains power link to the to Distribution Board ON BLOCK A DB	LM	20	200	4,000
ii)	16 mmØ for Mains power link to the to Distribution Board ON BLOCK B DB	LM	20	200	4,000
iii)	16 mmØ for Mains power link to the to Distribution Board ON BLOCK C DB'S	LM	50	200	10,000
iv)	16 mmØ for Mains power link to the Distribution Board ON BLOCK D DB	LM	10	200	2,000
v)	16 mmØ for Mains power link to the Distribution Board ON BLOCK E DB	LM	15	200	3,000
5	***METAL TRUNKING***				

(a)	100mmx50mmx2400mm; 2 compartment polished skirting metal trunking complete with all accessories for power points and LAN cabling.	LM	40	2,000	80,000
(b)	Corners for skirting metal trunking complete with all accessories.	No	10	1,000	10,000
6	Any other ITEM not included under the ELECTRICAL POWER DISTRIBUTION - that is necessary to complete this Electrical Installations as specified. (Provide Details Below)	Item	1	0	0
	SUB-TOTAL FOR BILL No 5.1 - ELECTRICAL POWER DISTRIBUTION - CARRIED FORWARD TO ELECTRICAL DISTRIBUTION COLLECTION PAGE				**320,000.00**
BILL NO 3 - ELECTRICAL DISTRIBUTION COLLECTION PAGE					
ITEM	**DESCRIPTION**				**BQ AMOUNT**
1	SUB-TOTAL FOR BILL No 3.1 - ELECTRICAL DISTRIBUTION BROUGHT FORWARD				438,000.00
	TOTAL FOR BILL No 5 -POWER DISTRIBUTION CARRIED FORWARD TO MAIN SUMMARY PAGE				**438,000.00**

BILL No 6 - MAIN LOW VOLTAGE SWITCHGEAR					
ITEM	**DESCRIPTION**	**UNIT**	**BQ QTY**	**Unit Cost (KSh)**	**Total Cost (Ksh)**
	ALL RATES SHALL BE INCLUSIVE OF CONTRACTUAL OBLIGATIONS, DUTIES AND GOVERNMENT TAXES				
	Indicated Product Brands are only a guide to the quality required				
	Supply, install, test, commission and hand over in proper working condition the following.				
1	250A TP Free Standing Type Tested, Modular, IP 44, Form 2B, Low Voltage main Switchboard, 250A TPN MCCB adjustable incomer with shunt trip, **to accomodate a 250A TP AMF panel, By-pass switch** and all outgoing feeders, 350A busbars, Instrumentation, earthing, labels. To be fabricated and assembled to **IEC** 60439-1 Standards from Heavy Gauge steel, powder coated finish in approved colour, floor mounted with entry cable provisions, lockable and fitted with a view glass for the meter position and space to accommodate the Electrical components detailed and specified in the main Schematic diagram.	Item	1	600,000	600,000
2	250A TPN MCCB adjustable incomer c/w shunt trip for incorporation in the Low Voltage Switchboard (LVSB) as MERLIN GERIN or equal and approved.	No	1	60,000	60,000

ITEM	**DESCRIPTION**	**UNIT**	**BQ QTY**	**Unit Cost (KSh)**	**Total Cost (Ksh)**
3	Allow for earthing of the LV Switchboard	Sum	1	10,000	10,000

5	100KVAr, 415V, 50Hz Upgradable Automatic Power Factor Capacitor Bank with a digital display of Power Factor, Capacitor Step and Set up information.	Item	1	200,000	200,000
6	Earthing for the APFC above.	Sum	1	5,000	5,000
7	Any other item necessary to complete the installation in this section as specified. (Provide Details below)	Sum		0	0
8	Multifunctional KPLC TP Meter at supply intake	No	0	50,000	0
	SUB-TOTAL FOR BILL No 6 - MAIN LOW VOLTAGE SWITCH GEAR CARRIED FORWARD TO THE MAIN SUMMARY PAGE				**875,000.00**

BILL No 7 - INCOMING ELECTRICITY SUPPLY					
ITEM	**DESCRIPTION**	**UNIT**	**BQ QTY**	**Unit Cost (KSh)**	**Total Cost (Ksh)**
	ALL RATES SHALL BE INCLUSIVE OF CONTRACTUAL OBLIGATIONS, DUTIES AND GOVERNMENT TAXES				
	Indicated Product Brands are only a guide to the quality required				
	Supply, install, test, commission and hand over in proper working condition the following.				
1	Allow a Prime Cost Sum (P.C) of KSh. 3,000,000.00 as Capital Contribution towards Kenya Power for Electricity supply line, Metering Channel and Connection Charges.	Item	1	3,000,000	3,000,000.00
2	Allow for attendance to Kenya Power (KP) with respect to the following:-				
(i)	- Application for New Electricity Supply line	Item	1	10,000	10,000.00
(ii)	- Follow up with KPLC on Quotation and new supply	Item	1	15,000	15,000.00
(iii)	- Liaising between the employer and KPLC with regards to signing of relevant contract documents and payment of requisite amounts for new electricity supply line	Item	1	10,000	10,000.00
3	Trenching, sifting, tiling and backfilling above 2x100mm diameter Heavy Gauge PVC Duct with concrete Surround under road/along boundary wall	LM	20	1,000	20,000.00
4	Any other item necessary to complete the installation in this section as specified. (Provide Details below)	Sum	1	0	0.00
5	Standard cable markers	No.	0	400	0.00
	SUB-TOTAL FOR BILL No 7 - INCOMING ELECTRICITY SUPPLY CARRIED FORWARD TO THE MAIN SUMMARY PAGE				**3,055,000.00**

PRICES SUMMARY BILL 1 TO 7		
ITEM	**DESCRIPTION**	**BQ AMOUNT**
1	TOTAL FOR BILL No 2 - LIGHTING INSTALLATIONS brought forward to Main Summary Page	95,050.00
2	TOTAL FOR BILL No 3 - POWER INSTALLATIONS brought forward to Main Summary Page	402,200.00
3	TOTAL FOR BILL No 5 - ELECTRICAL POWER DISTRIBUTION brought forward to Main Summary Page	438,000.00
4	TOTAL FOR BILL No 6 - MAIN L.V SWITCHGEAR brought forward to Main Summary Page	875,000.00
5	TOTAL FOR BILL No 7 - ELECTRICAL INCOMING POWER brought forward to Main Summary Page	3,055,000.00
6	ALLOW KSH. 300,000.00 CONTINGENCY SUM to be expended at the discretion of the Engineer.	**300,000.00**
	TOTAL CARRIED TO INTERIM PAYMENT CERTIFICATE INCLUSIVE OF DUTIES AND TAXES, Ksh	**5,165,250.00**

ANALYSIS OF POSSIBLE RENEWABLE ENERGY SOLUTIONS					
ITEM	**DESCRIPTION**	**UNIT**	**BQ QTY**	**UNIT COST (Ksh)**	**TOTAL COST (Ksh)**
	Indicated Product Brands are only a guide to the quality required				
	Supply, install, test, commission and hand over in proper working condition				

	the following.				
	SOLAR				
1	48V ,300Ah Solar Batteries as Taico 15 kWh 48V Lithium Battery | TKRB-1500 51.2V 300Ah 15.36kW	No	8	300,000	2,400,000
2	550 W Solar Panel	No	44	18,800	827,200
3	15Kw Solar Inverter as TBB Matrix II 15 kVA 48V Hybrid Solar Inverter & WiFi Dongle | Low Frequency	No	4	500,000	2,000,000
	TOTAL FOR BILL No 5 – ANALYSIS OF RENEWABLE ENERGY brought forward to Main Summary Page				**5,227,200**

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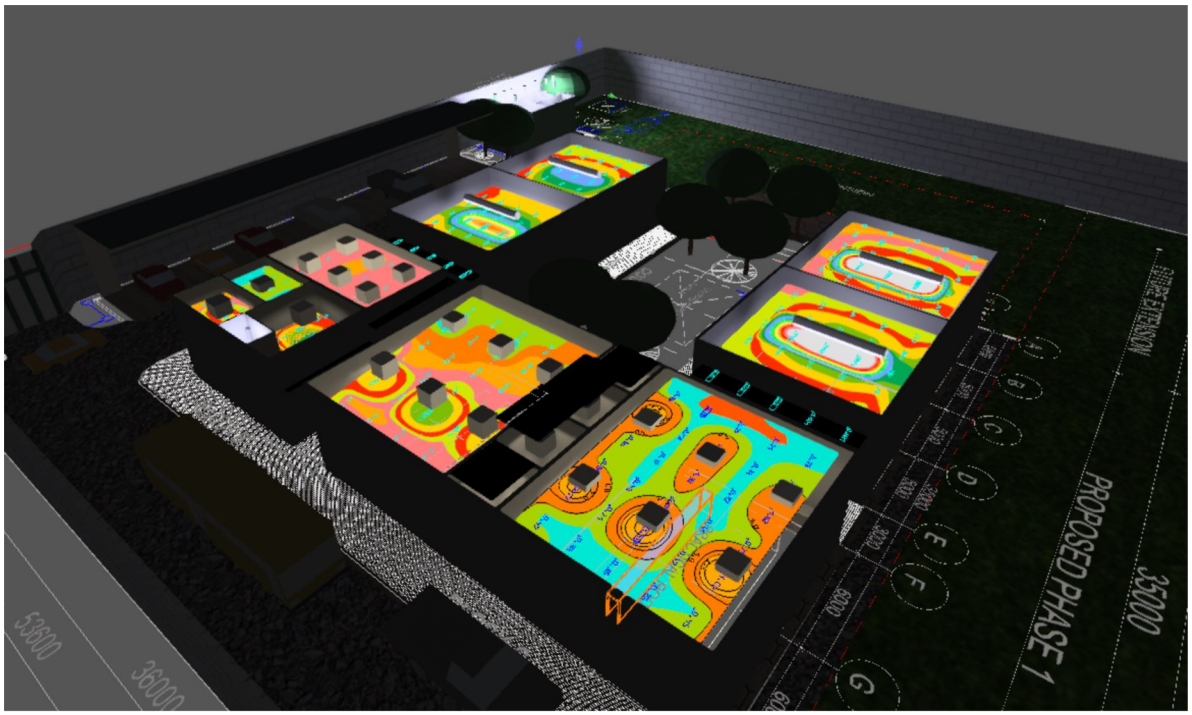
LEAD ARCHITECT

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LEAD LIGHTING DESIGNER



Description

The DIALux lighting schedule for the HADU Vocational Training Centre provides a comprehensive overview of all luminaires specified within the project. The schedule consolidates technical, electrical, and photometric data for each fixture type used across classrooms, practical rooms, administrative spaces, and external areas.

The selected luminaires are high-efficiency LED fittings designed to deliver uniform illumination while minimizing energy consumption. Photometric data from certified manufacturer files (IES/LDT) was used to simulate and verify lighting performance, ensuring accurate representation of real-world conditions.

The lighting design achieves appropriate illuminance levels for educational and vocational environments, with adequate average lux values and acceptable uniformity ratios across task and circulation areas. Particular emphasis has been placed on workbench and practical zones to ensure visual clarity, safety, and compliance with recommended lighting standards for learning facilities.

From an electrical perspective, the lighting schedule details the installed wattage per luminaire, quantity per space, and total connected lighting load. This information supports proper circuit sizing, distribution board configuration, and overall energy planning.

The system has been designed to optimize lighting performance while maintaining energy efficiency, long service life, and reduced maintenance requirements. Overall, the lighting schedule confirms that the proposed

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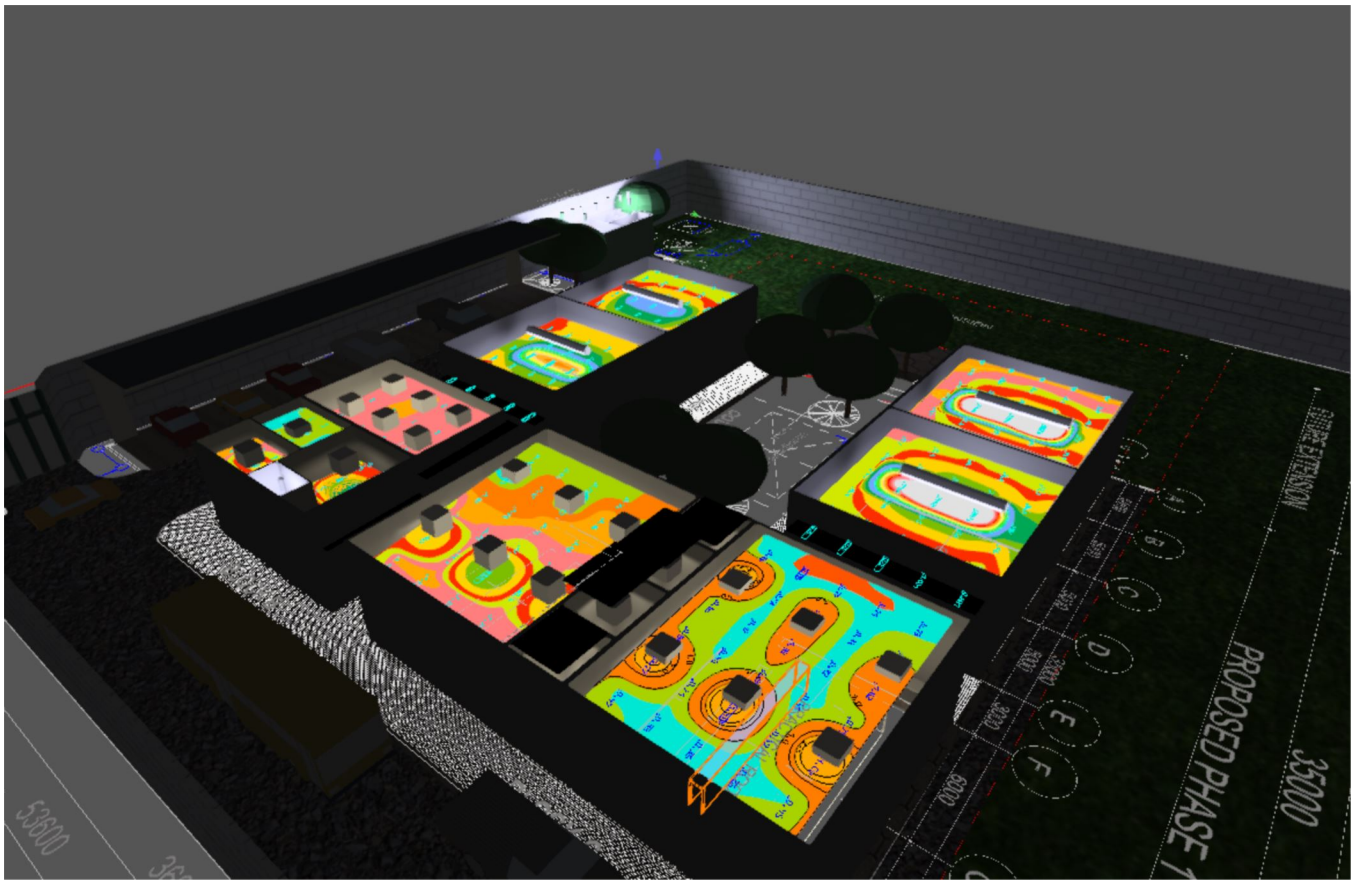
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installation meets functional, technical, and operational requirements for the facility.

Images

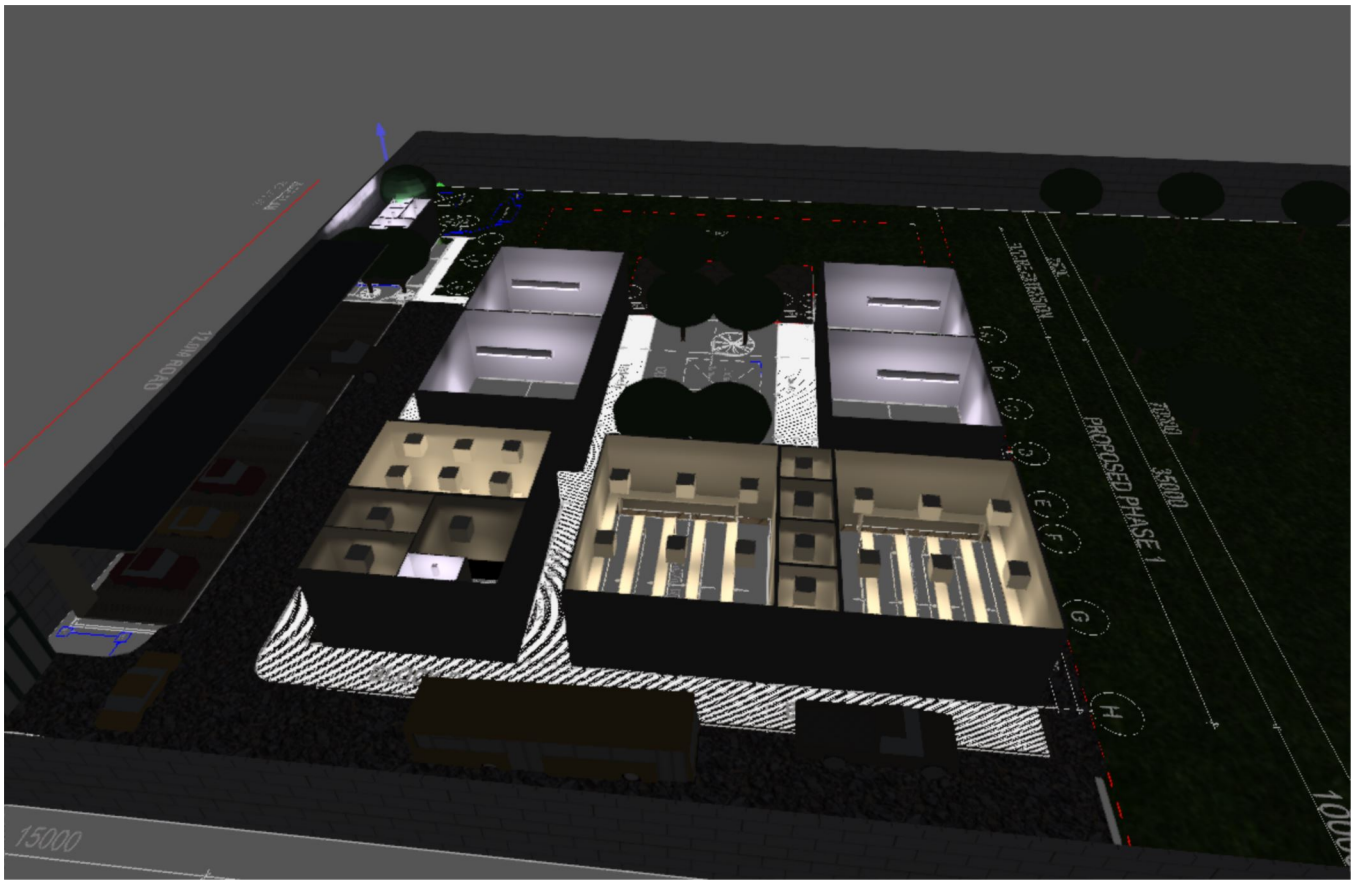


SITE PLAN 3D EXPLODED LIGHTING VIEW

Full photometric view of HADU vocational Centre

This 3D exploded photometric view illustrates the overall lighting distribution across the HADU Vocational Training Centre. The simulation highlights illuminance levels using colour contour mapping, where warmer colours (red/orange) indicate higher lux levels and cooler colours (green/blue) represent lower illuminance areas. The view demonstrates uniform light coverage across classrooms, practical rooms, circulation paths, and external spaces, ensuring compliance with recommended lighting standards for educational facilities.

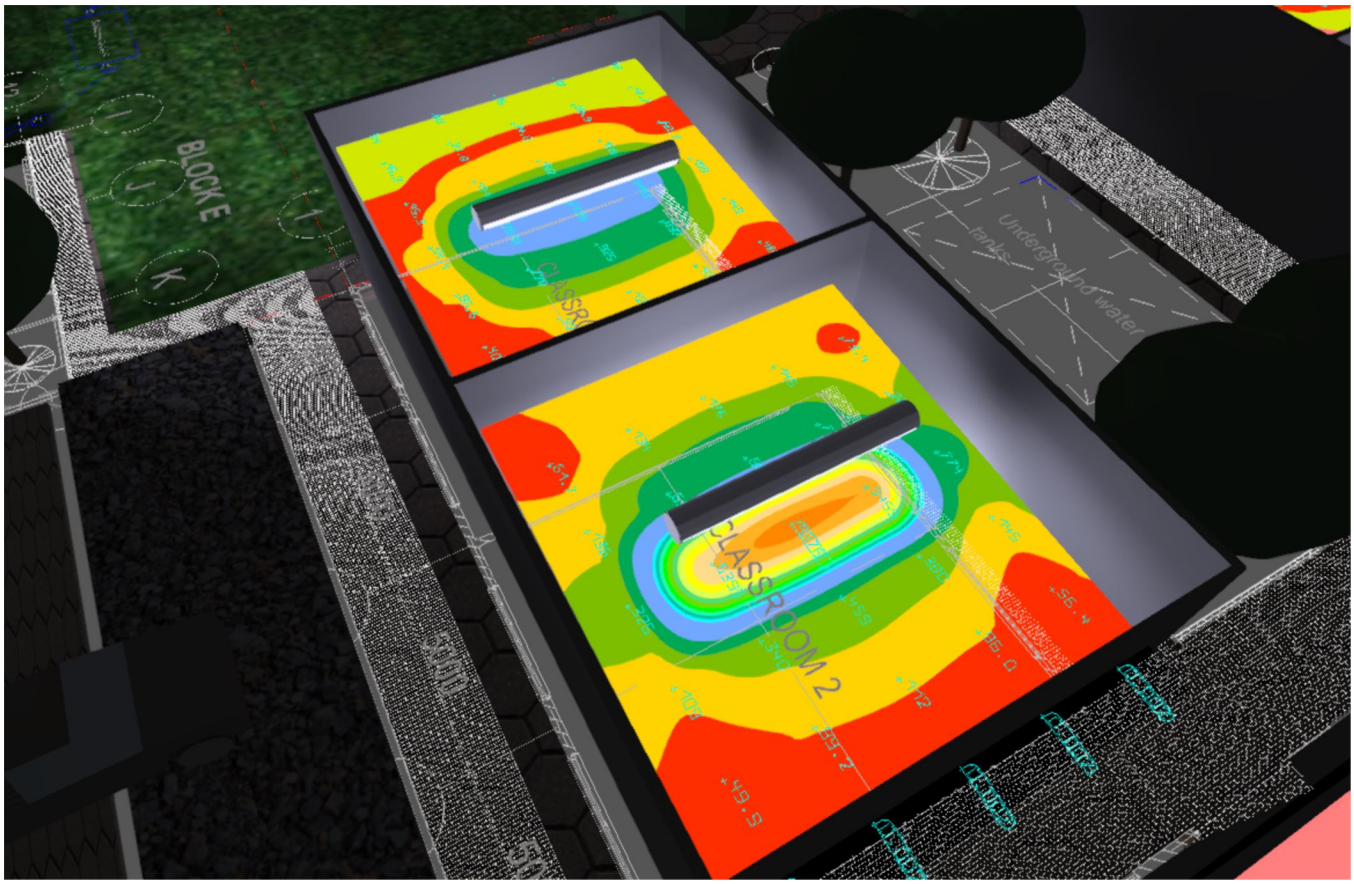
Images



EXPLODED RENDER

This exploded architectural render provides a comprehensive visualization of the building layout and lighting arrangement. The model displays the spatial organization of classrooms, practical rooms, and support spaces, while illustrating luminaire placement within each zone. The rendering assists in evaluating fixture positioning, ceiling integration, and overall visual comfort within the learning environment.

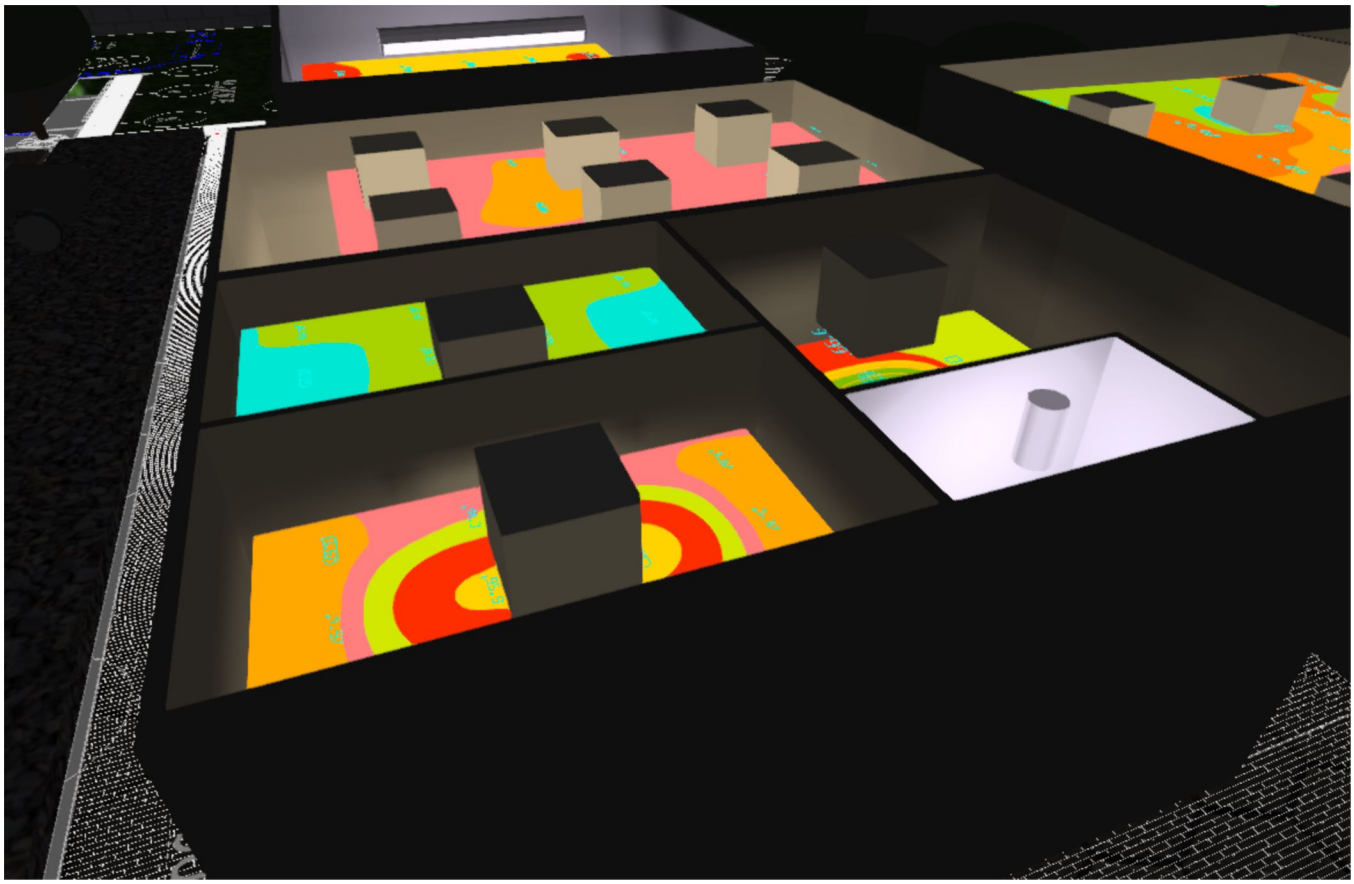
Images



BLOCK A

This photometric view presents the lighting analysis for Classrooms 1 and 2. The contour lines and colour gradients represent measured illuminance distribution across work surfaces and circulation areas. The results demonstrate adequate average lux levels, acceptable uniformity ratios, and minimized dark spots, ensuring suitable lighting conditions for academic and practical activities.

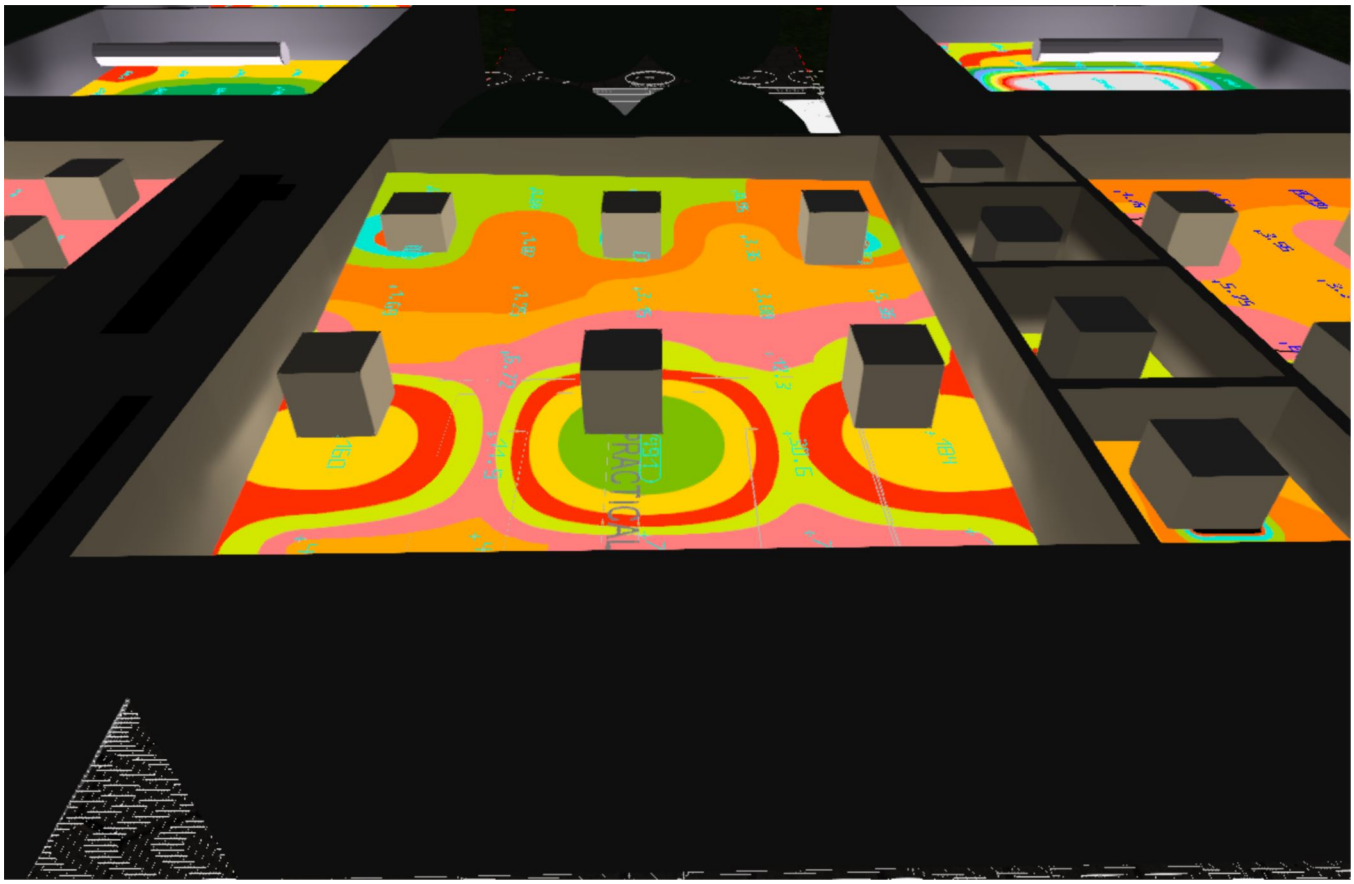
Images



BLOCK B

This lighting simulation for the administrative office demonstrates appropriate illuminance levels for desk-based tasks and computer usage. The design provides uniform lighting distribution, reducing shadows and glare while maintaining visual comfort in accordance with office lighting recommendations.

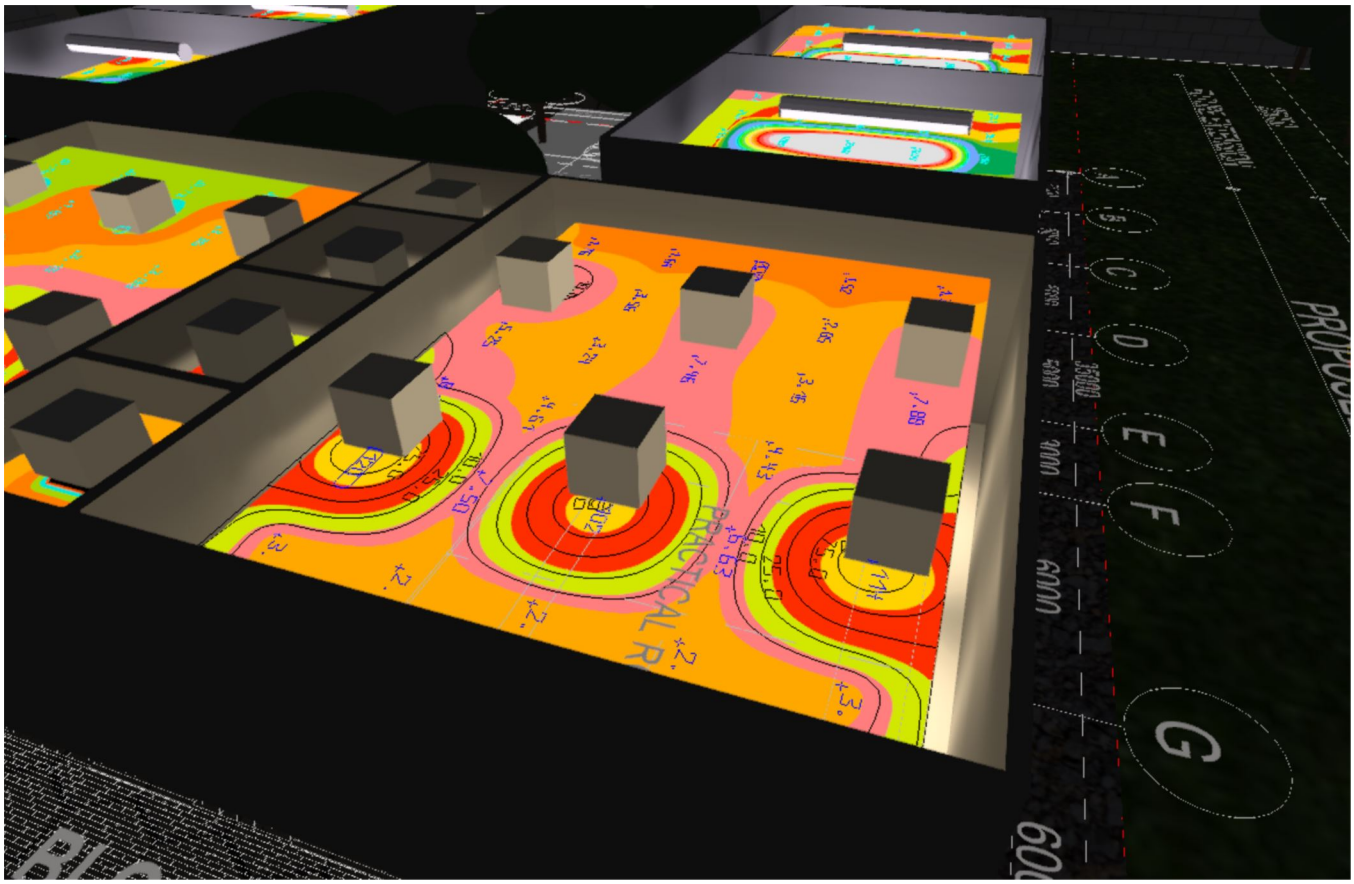
Images



PRACTICAL ROOM 1

This photometric representation illustrates the lighting distribution within Practical Room 1. The analysis verifies balanced light coverage across operational zones, with adequate task lighting levels for hands-on instructional activities. The design minimizes glare while maintaining consistent illuminance throughout the workspace.

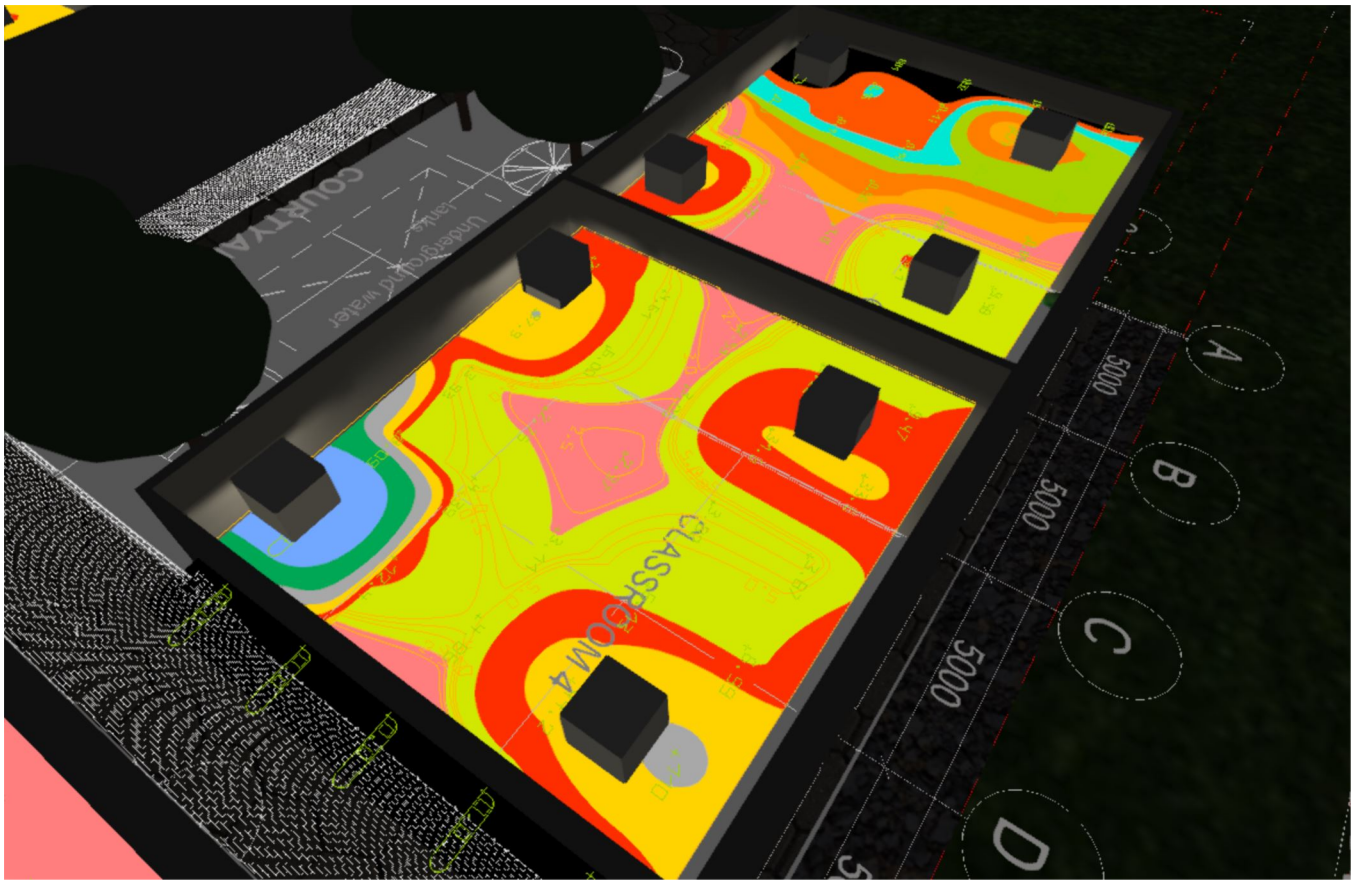
Images



PRACTICAL ROOM 2

This image shows the simulated lighting performance within Practical Room 2. The contour mapping confirms sufficient illuminance on workbenches and equipment areas, supporting detailed technical tasks. Light uniformity across the space ensures visual comfort, safety, and compliance with recommended lux levels for vocational training environments.

Images



BLOCK D

This photometric view presents the lighting analysis for Classrooms 3 and 4. The contour lines and colour gradients represent measured illuminance distribution across work surfaces and circulation areas. The results demonstrate adequate average lux levels, acceptable uniformity ratios, and minimized dark spots, ensuring suitable lighting conditions for academic and practical activities.

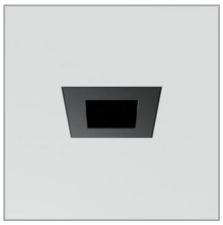
Luminaire list

Φ_{total}	P_{total}	Luminous efficacy
27841 lm	553.6 W	50.3 lm/W

pcs.	Manufacturer	Article No.	Article name	P	Φ	Luminous efficacy
41	formalighting	RP94MC31-902715	Spotify 100 Adjustable Square - Trimless	12.6 W	601 lm	47.7 lm/W
10	V-TAC	8043	V-TAC LED Bulb - SAMSUNG CHIP 3.7W E14 P45 Plastic 4000K	3.7 W	320 lm	86.5 lm/W

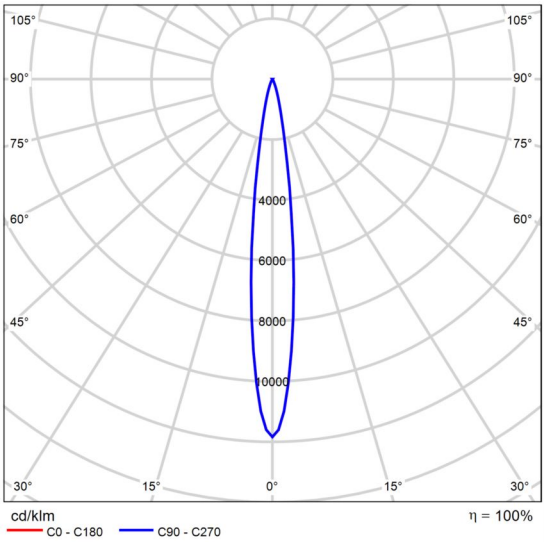
Product data sheet

formalighting - Spotify 100 Adjustable Square - Trimless

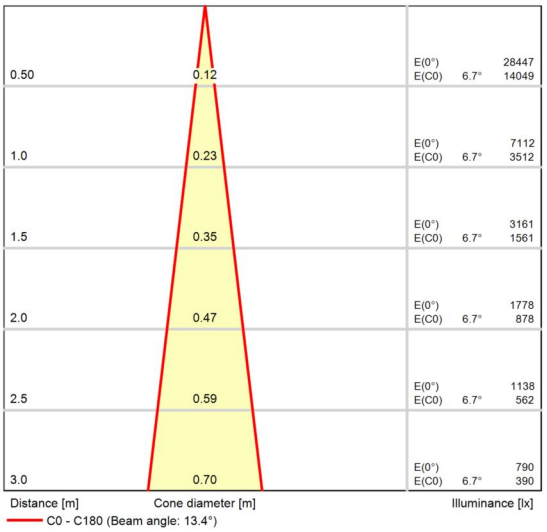


Article No.	RP94MC31-902715
P	12.6 W
Φ_{Lamp}	601 lm
$\Phi_{Luminaire}$	601 lm
η	100.02 %
Luminous efficacy	47.7 lm/W
CCT	2700 K
CRI	80

Following formalighting's Eco Design philosophy, Spotify Adjustable Square provides high visual comfort thanks to a deep recessed LED Engine.
Available in 2 sizes 77x 77; and 100 x 100, Trim & Trimless versions. with modular and tool-free LED-engine and optical accessories site changeable options. Tiltable vertically 0-32° tilt and rotatable horizontally 355°.



Polar LDC



Cone diagram

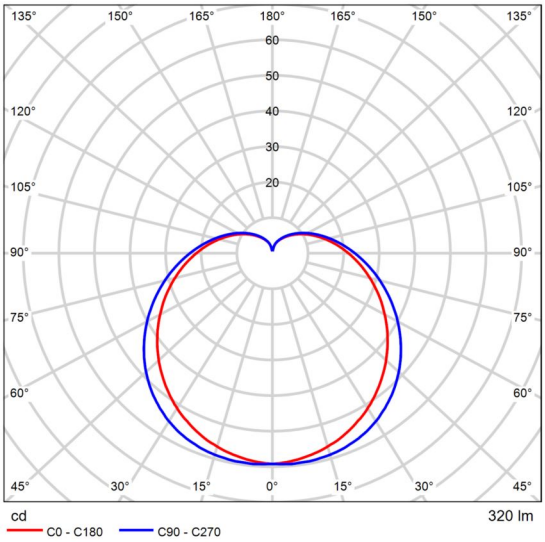
Product data sheet

V-TAC - V-TAC LED Bulb - SAMSUNG CHIP 3.7W E14 P45 Plastic 4000K



Article No.	8043
P	3.7 W
Φ_{Lamp}	–
$\Phi_{Luminaire}$	320 lm
η	–
Luminous efficacy	86.5 lm/W
CCT	4000 K
CRI	80

V-TAC LED Bulb - SAMSUNG CHIP 3.7W E14 P45 Plastic 4000K



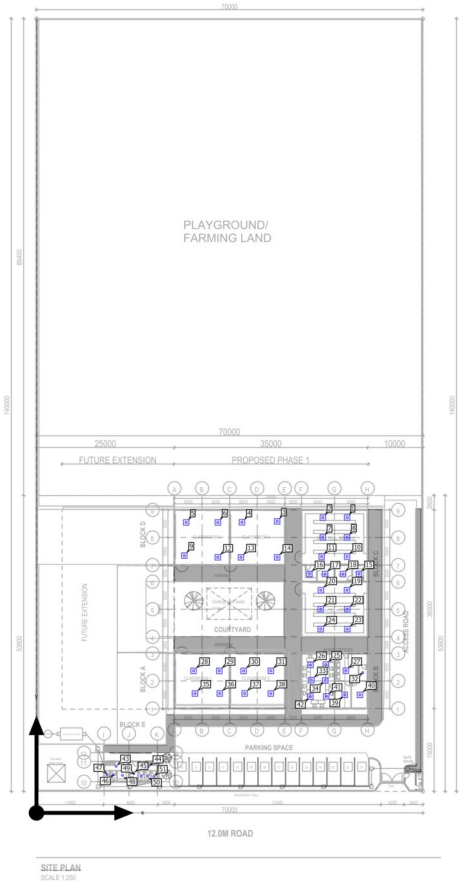
Polar LDC

Glare evaluation according to UGR

ρ Ceiling		70	70	50	50	30	70	70	50	50	30
ρ Walls		50	30	50	30	30	50	30	50	30	30
ρ Floor		20	20	20	20	20	20	20	20	20	20
Room size X	Y	Viewing direction at right angles to lamp axis					Viewing direction parallel to lamp axis				
2H	2H	24.2	25.4	24.8	26.0	26.7	24.7	25.9	25.3	26.5	27.2
	3H	26.7	27.8	27.3	28.4	29.2	27.3	28.4	27.9	29.0	29.7
	4H	28.1	29.1	28.7	29.8	30.5	28.7	29.7	29.3	30.4	31.1
	6H	29.6	30.6	30.3	31.2	32.0	30.2	31.2	30.8	31.8	32.6
	8H	30.4	31.4	31.1	32.1	32.9	31.0	32.0	31.7	32.6	33.4
	12H	31.4	32.3	32.1	33.0	33.8	32.0	32.9	32.7	33.6	34.4
4H	2H	25.2	26.2	25.8	26.9	27.6	25.5	26.6	26.2	27.2	28.0
	3H	27.9	28.8	28.5	29.5	30.3	28.3	29.2	29.0	29.9	30.7
	4H	29.4	30.2	30.1	30.9	31.7	29.9	30.7	30.6	31.4	32.2
	6H	31.1	31.8	31.8	32.5	33.4	31.6	32.3	32.3	33.0	33.9
	8H	32.0	32.7	32.8	33.4	34.3	32.5	33.2	33.3	34.0	34.8
	12H	33.1	33.7	33.9	34.5	35.4	33.6	34.3	34.4	35.0	35.9
8H	4H	30.1	30.8	30.8	31.5	32.4	30.4	31.1	31.2	31.9	32.7
	6H	32.0	32.6	32.8	33.4	34.3	32.5	33.0	33.2	33.8	34.7
	8H	33.2	33.7	33.9	34.5	35.4	33.6	34.1	34.4	34.9	35.8
	12H	34.5	34.9	35.3	35.7	36.7	34.9	35.4	35.7	36.2	37.1
12H	4H	30.2	30.9	31.0	31.6	32.5	30.6	31.2	31.3	32.0	32.8
	6H	32.3	32.8	33.1	33.6	34.5	32.7	33.2	33.5	34.0	34.9
	8H	33.6	34.0	34.4	34.8	35.8	34.0	34.4	34.7	35.2	36.1
Variation of the observer position for the luminaire distances S											
S = 1.0H		+0.1 / -0.1					+0.1 / -0.1				
S = 1.5H		+0.3 / -0.3					+0.3 / -0.3				
S = 2.0H		+0.4 / -0.5					+0.4 / -0.5				
Standard table		BK12					BK12				
Correction summand		18.3					18.7				
Corrected glare indices referring to 320lm Total luminous flux											

UGR diagram (SHR: 0.25)

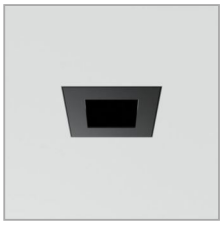
Luminaire layout plan

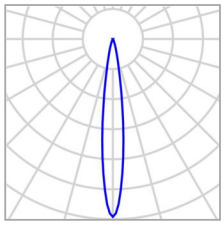


Site 1

Luminaire layout plan







Manufacturer	formalighting	P	12.6 W
Article No.	RP94MC31-902715	$\Phi_{Luminaire}$	601 lm
Article name	Spotify 100 Adjustable Square - Trimless		
Fitting	1x Spotify 100 Adjustable Square - Trimless		

Individual luminaires

X	Y	Mounting height	Luminaire
7.387 m	7.025 m	0.750 m	1
6.809 m	7.022 m	0.750 m	2
5.723 m	6.932 m	0.750 m	3
4.885 m	6.911 m	0.750 m	4
3.561 m	6.907 m	0.750 m	5
4.312 m	6.907 m	0.750 m	6
6.807 m	6.550 m	0.750 m	7
7.383 m	6.550 m	0.750 m	8
3.515 m	6.120 m	0.750 m	9
7.387 m	6.094 m	0.750 m	10
6.781 m	6.094 m	0.750 m	11
4.312 m	6.077 m	0.750 m	12

Site 1

Luminaire layout plan

X	Y	Mounting height	Luminaire
4.860 m	6.077 m	0.750 m	13
5.723 m	6.077 m	0.750 m	14
7.668 m	5.681 m	0.750 m	15
6.489 m	5.680 m	0.750 m	16
6.861 m	5.680 m	0.750 m	17
7.287 m	5.680 m	0.750 m	18
7.383 m	5.295 m	0.750 m	19
6.761 m	5.295 m	0.750 m	20
6.757 m	4.844 m	0.750 m	21
7.391 m	4.844 m	0.750 m	22
7.391 m	4.369 m	0.750 m	23
6.757 m	4.369 m	0.750 m	24
6.878 m	3.526 m	0.750 m	25
6.516 m	3.517 m	0.750 m	26
7.355 m	3.383 m	0.750 m	27
3.735 m	3.378 m	0.750 m	28
4.346 m	3.378 m	0.750 m	29
4.924 m	3.378 m	0.750 m	30
5.571 m	3.378 m	0.750 m	31
6.537 m	3.157 m	0.750 m	33
6.863 m	3.149 m	0.750 m	34
3.768 m	2.839 m	0.750 m	35
4.357 m	2.839 m	0.750 m	36
4.948 m	2.839 m	0.750 m	37

Site 1

Luminaire layout plan

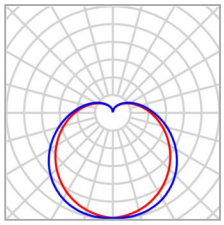
X	Y	Mounting height	Luminaire
5.571 m	2.839 m	0.750 m	38
7.328 m	2.806 m	0.750 m	39
7.690 m	2.806 m	0.750 m	40
6.889 m	2.771 m	0.750 m	41
6.512 m	2.767 m	0.750 m	42

Site 1

Luminaire layout plan







Manufacturer	V-TAC	P	3.7 W
Article No.	8043	$\Phi_{Luminaire}$	320 lm
Article name	V-TAC LED Bulb - SAMSUNG CHIP 3.7W E14 P45 Plastic 4000K		
Fitting	1x V-TAC LED Bulb - SAMSUNG CHIP 3.7W E14 P45 Plastic 4000K		

Individual luminaires

X	Y	Mounting height	Luminaire
7.757 m	3.335 m	0.750 m	32
1.893 m	1.107 m	0.500 m	43
2.661 m	1.101 m	0.500 m	44
2.310 m	0.944 m	0.500 m	45
1.832 m	0.908 m	0.500 m	46
1.679 m	0.901 m	0.500 m	47
2.045 m	0.895 m	0.500 m	48
2.489 m	0.879 m	0.500 m	49
2.640 m	0.879 m	0.500 m	50
2.792 m	0.865 m	0.500 m	51

Site 1

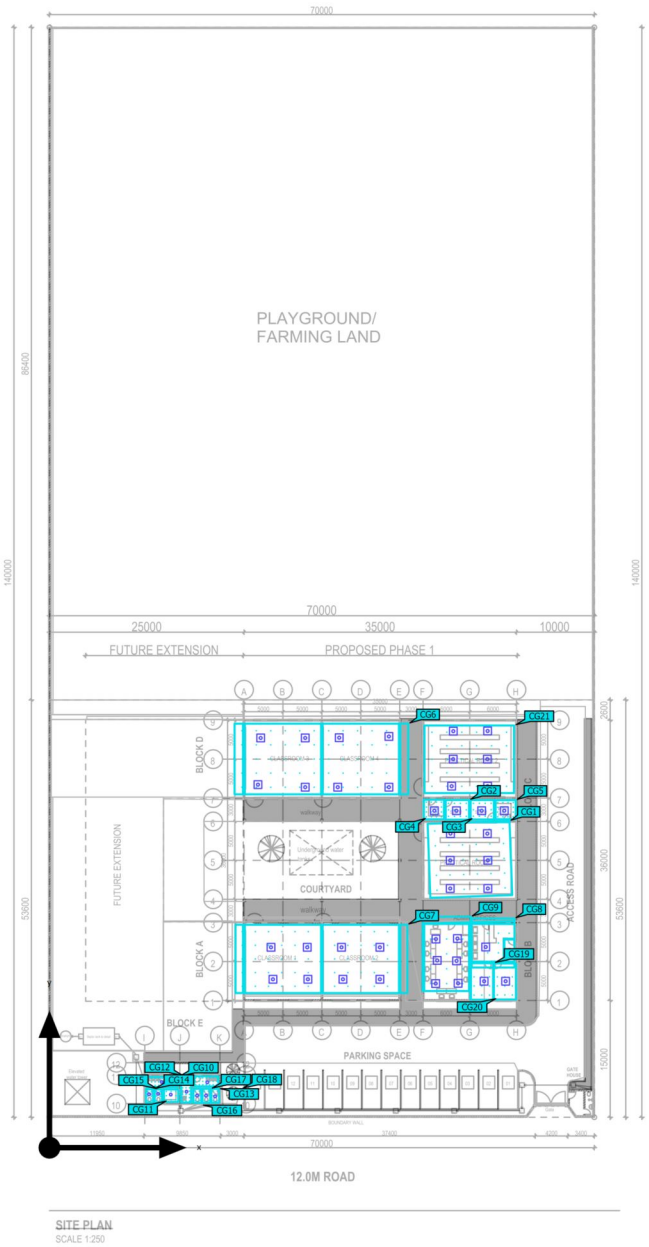
Luminaire list

Φ_{total}	P_{total}	Luminous efficacy
27841 lm	553.6 W	50.3 lm/W

pcs.	Manufacturer	Article No.	Article name	P	Φ	Luminous efficacy
41	formalighting	RP94MC31-902715	Spotify 100 Adjustable Square - Trimless	12.6 W	601 lm	47.7 lm/W
10	V-TAC	8043	V-TAC LED Bulb - SAMSUNG CHIP 3.7W E14 P45 Plastic 4000K	3.7 W	320 lm	86.5 lm/W

Site 1 (Light scene 1)

Calculation objects



Site 1 (Light scene 1)

Calculation objects

Calculation surfaces

Properties	Ē	E_{min}	E_{max}	U_o (g_1)	g_2	Index
CLASSROOM 1 AND 2 Perpendicular illuminance Height: 0.547 m	120 lx	0.000 lx	4801 lx	0.00	0.00	CG7
CLASSROOM 3 AND 4 Perpendicular illuminance Height: 0.605 m	51.8 lx	0.00 lx	2574 lx	0.00	0.00	CG6
OFFICE 1 Perpendicular illuminance Height: 0.671 m	0.69 lx	0.29 lx	2.03 lx	0.42	0.14	CG19
OFFICE 2 Perpendicular illuminance Height: 0.606 m	27.1 lx	3.78 lx	129 lx	0.14	0.029	CG20
PRACTICAL ROOM 1 Perpendicular illuminance Height: 0.619 m	33.1 lx	0.00 lx	484 lx	0.00	0.00	CG1
PRACTICAL ROOM 2 Perpendicular illuminance Height: 0.619 m	15.9 lx	1.59 lx	123 lx	0.10	0.013	CG21
PREP 1 Perpendicular illuminance Height: 0.532 m	819 lx	36.6 lx	8850 lx	0.045	0.004	CG4
PREP ROOM 2 Perpendicular illuminance Height: 0.649 m	2.08 lx	0.00 lx	3.29 lx	0.00	0.00	CG5
PWD WASHROOM Perpendicular illuminance Height: 0.000 m	261 lx	16.3 lx	318 lx	0.062	0.051	CG12
RECEPTION Perpendicular illuminance Height: 0.480 m	268 lx	4.54 lx	1462 lx	0.017	0.003	CG8
SINK AREA Perpendicular illuminance Height: 0.000 m	251 lx	181 lx	314 lx	0.72	0.58	CG10

Site 1 (Light scene 1)

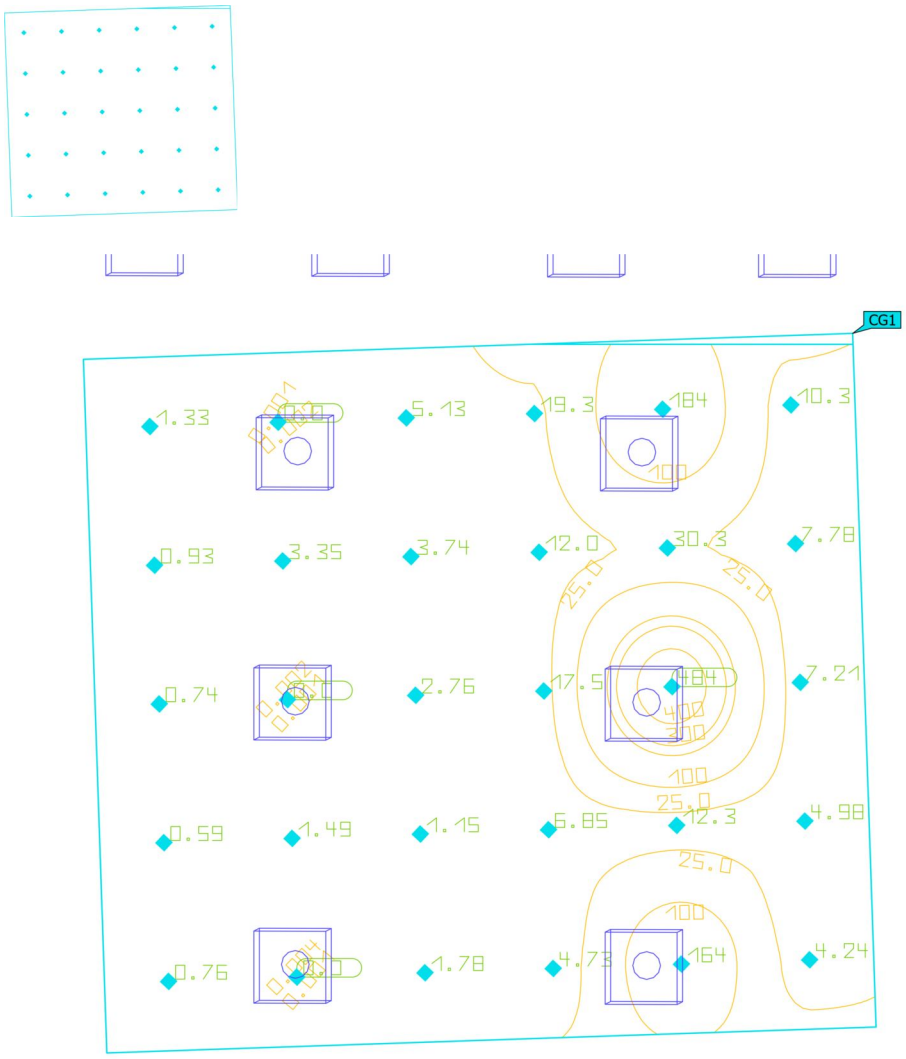
Calculation objects

SINK AREA 2 Perpendicular illuminance Height: 0.000 m	274 lx	214 lx	318 lx	0.78	0.67	CG13
STAFFROOM Perpendicular illuminance Height: 0.635 m	7.79 lx	5.64 lx	10.4 lx	0.72	0.54	CG9
STORE 1 Perpendicular illuminance Height: 0.558 m	537 lx	22.6 lx	5456 lx	0.042	0.004	CG2
STORE 2 Perpendicular illuminance Height: 0.605 m	85.1 lx	8.75 lx	566 lx	0.10	0.015	CG3
URINAL AREA Perpendicular illuminance Height: 0.000 m	303 lx	284 lx	331 lx	0.94	0.86	CG11
WASHROOM 1 Perpendicular illuminance Height: 0.000 m	214 lx	22.8 lx	314 lx	0.11	0.073	CG14
WASHROOM 2 Perpendicular illuminance Height: 0.000 m	214 lx	24.0 lx	310 lx	0.11	0.077	CG15
WASHROOM 3 Perpendicular illuminance Height: 0.500 m	4.13 lx	2.77 lx	5.75 lx	0.67	0.48	CG16
WASHROOM 4 Perpendicular illuminance Height: 0.000 m	210 lx	21.3 lx	308 lx	0.10	0.069	CG17
WASHROOM 5 Perpendicular illuminance Height: 0.500 m	0.47 lx	0.092 lx	1.50 lx	0.20	0.061	CG18

Utilisation profile: DIALux presetting (5.1.4 Standard (outdoor transportation area))

Site 1 (Light scene 1)

PRACTICAL ROOM 1

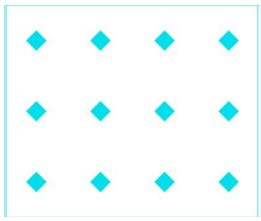


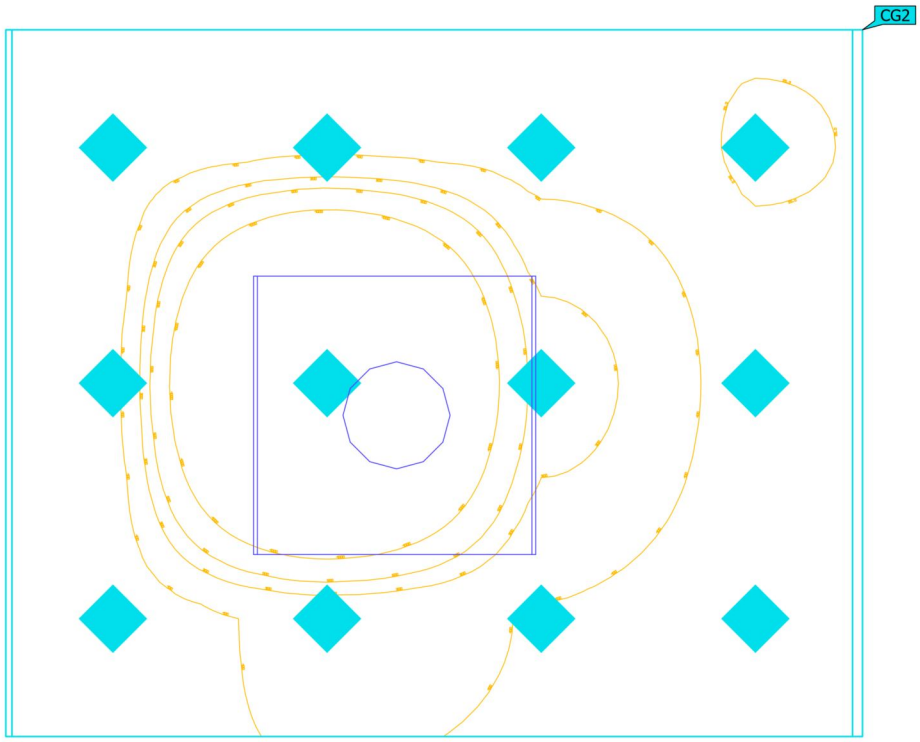
Properties	Ē	E_{min}	E_{max}	U_o (g_1)	g_2	Index
PRACTICAL ROOM 1 Perpendicular illuminance Height: 0.619 m	33.1 lx	0.00 lx	484 lx	0.00	0.00	CG1

Utilisation profile: DIALux presetting (5.1.4 Standard (outdoor transportation area))

Site 1 (Light scene 1)

STORE 1



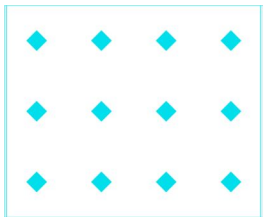


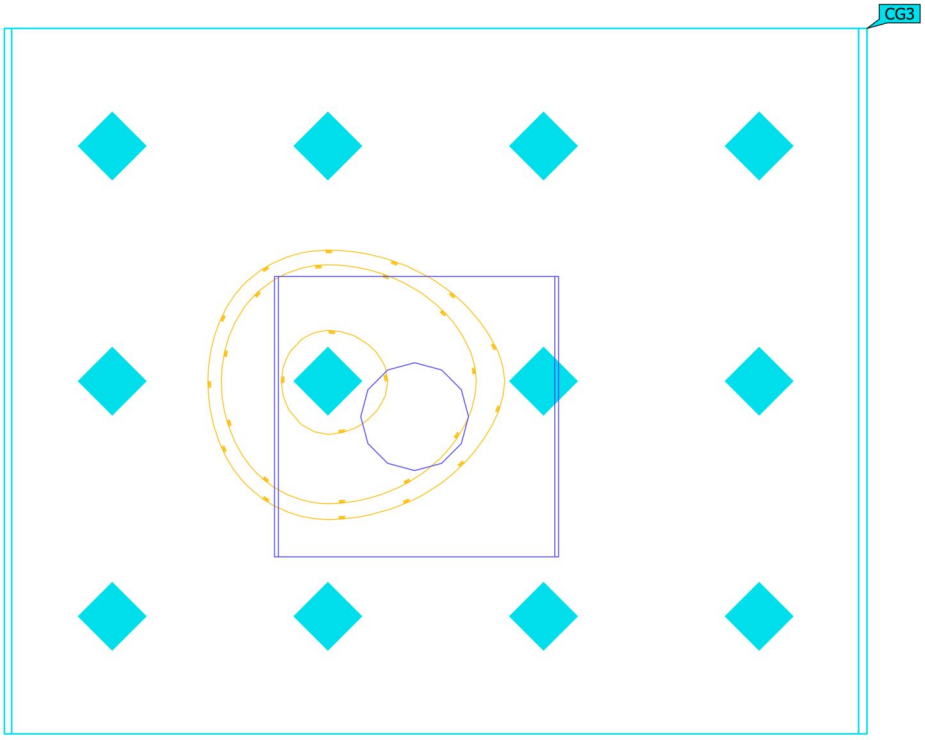
Properties	Ē	E_{min}	E_{max}	U_o (g_1)	g_2	Index
STORE 1 Perpendicular illuminance Height: 0.558 m	537 lx	22.6 lx	5456 lx	0.042	0.004	CG2

Utilisation profile: DIALux presetting (5.1.4 Standard (outdoor transportation area))

Site 1 (Light scene 1)

STORE 2



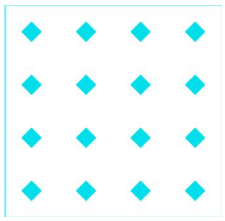


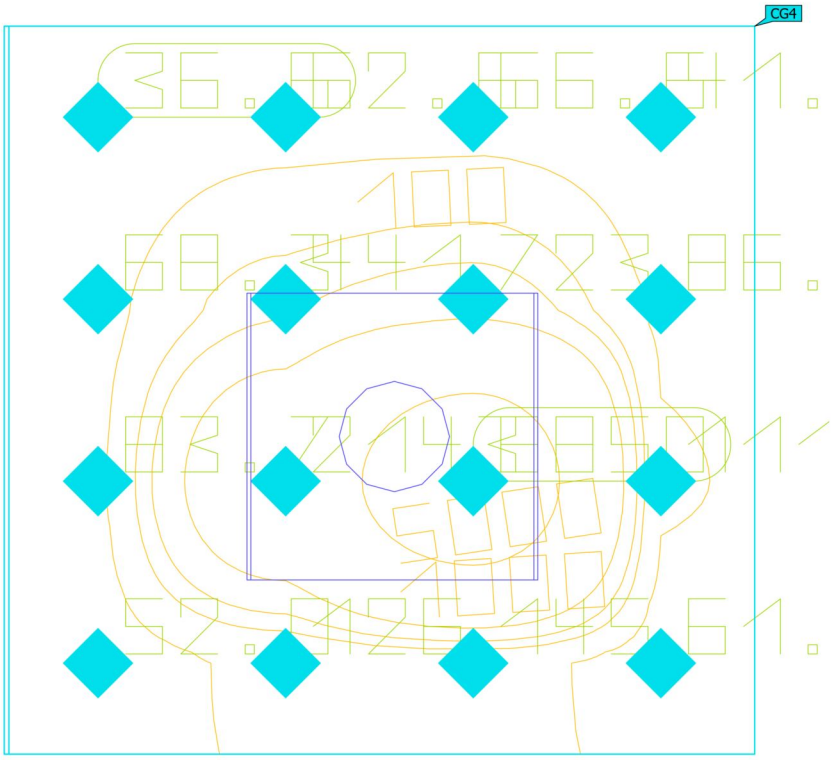
Properties	Ē	E_{min}	E_{max}	U_o (g_1)	g_2	Index
STORE 2 Perpendicular illuminance Height: 0.605 m	85.1 lx	8.75 lx	566 lx	0.10	0.015	CG3

Utilisation profile: DIALux presetting (5.1.4 Standard (outdoor transportation area))

Site 1 (Light scene 1)

PREP 1



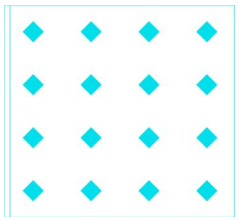


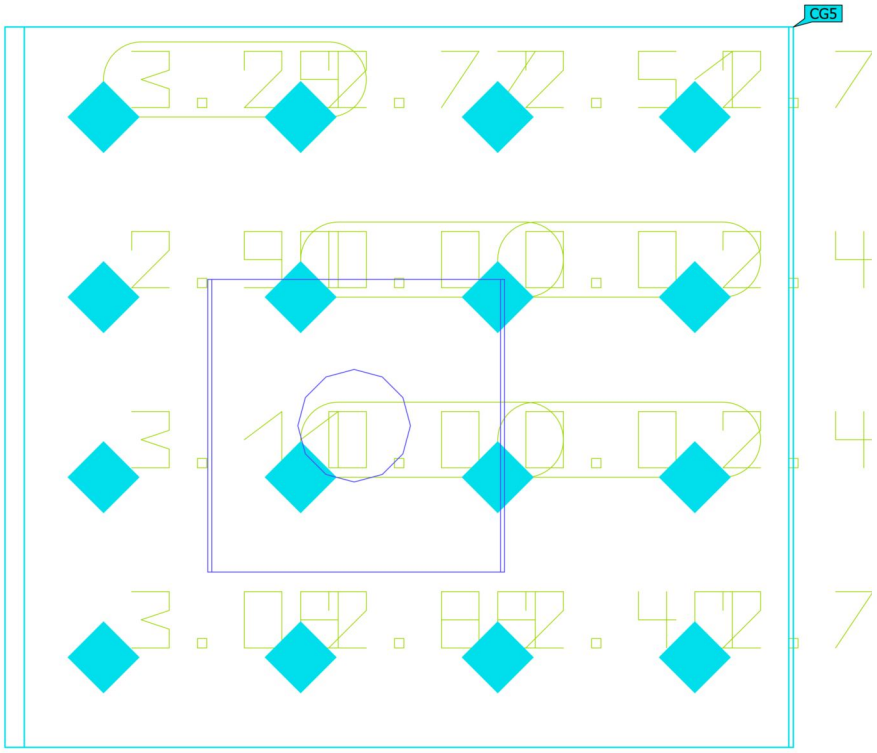
Properties	Ē	E_{min}	E_{max}	U_o (g_1)	g_2	Index
PREP 1 Perpendicular illuminance Height: 0.532 m	819 lx	36.6 lx	8850 lx	0.045	0.004	CG4

Utilisation profile: DIALux presetting (5.1.4 Standard (outdoor transportation area))

Site 1 (Light scene 1)

PREP ROOM 2



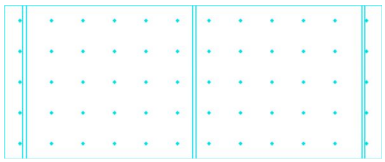


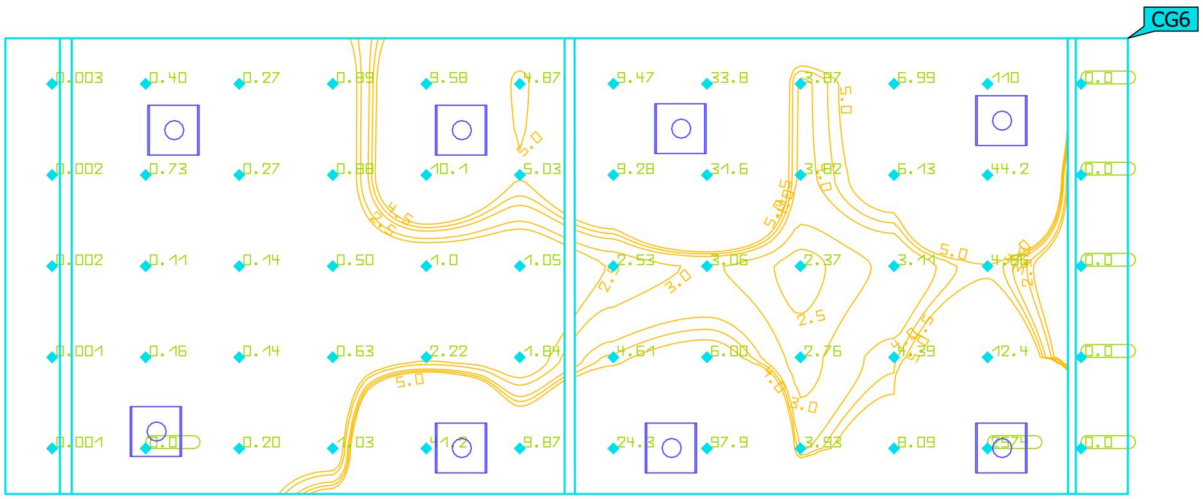
Properties	Ē	E_{min}	E_{max}	U_o (g_1)	g_2	Index
PREP ROOM 2 Perpendicular illuminance Height: 0.649 m	2.08 lx	0.00 lx	3.29 lx	0.00	0.00	CG5

Utilisation profile: DIALux presetting (5.1.4 Standard (outdoor transportation area))

Site 1 (Light scene 1)

CLASSROOM 3 AND 4



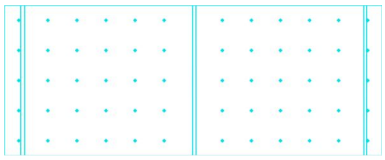


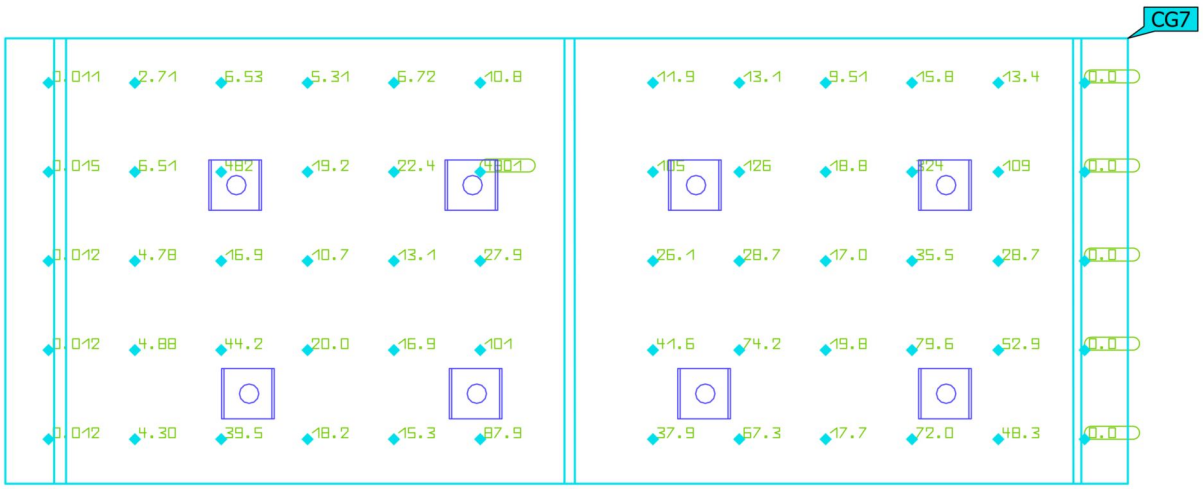
Properties	Ē	E_{min}	E_{max}	U_o (g_1)	g_2	Index
CLASSROOM 3 AND 4 Perpendicular illuminance Height: 0.605 m	51.8 lx	0.00 lx	2574 lx	0.00	0.00	CG6

Utilisation profile: DIALux presetting (5.1.4 Standard (outdoor transportation area))

Site 1 (Light scene 1)

CLASSROOM 1 AND 2



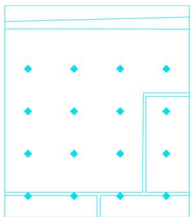


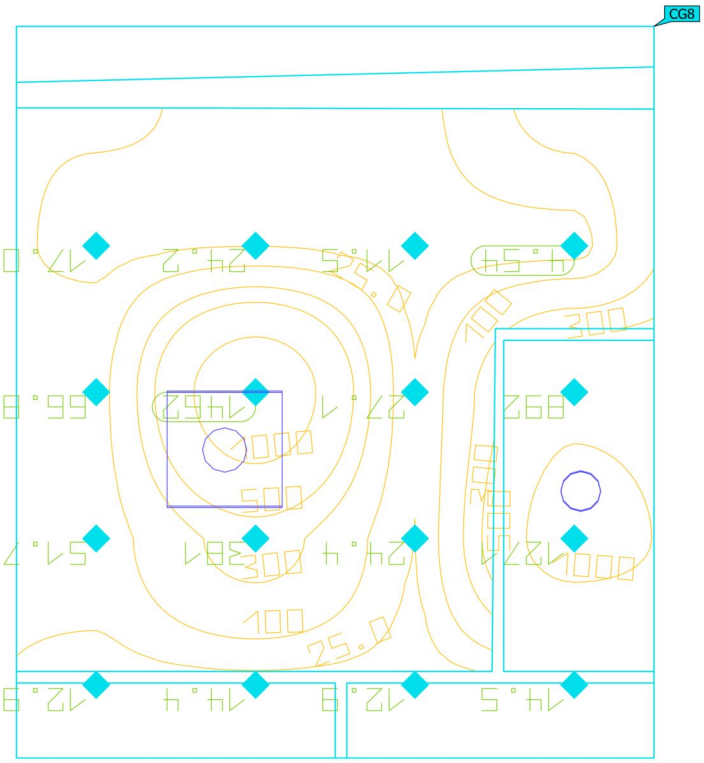
Properties	Ē	E_{min}	E_{max}	U_o (g_1)	g_2	Index
CLASSROOM 1 AND 2 Perpendicular illuminance Height: 0.547 m	120 lx	0.000 lx	4801 lx	0.00	0.00	CG7

Utilisation profile: DIALux presetting (5.1.4 Standard (outdoor transportation area))

Site 1 (Light scene 1)

RECEPTION



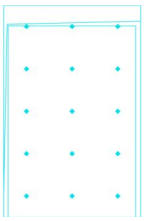


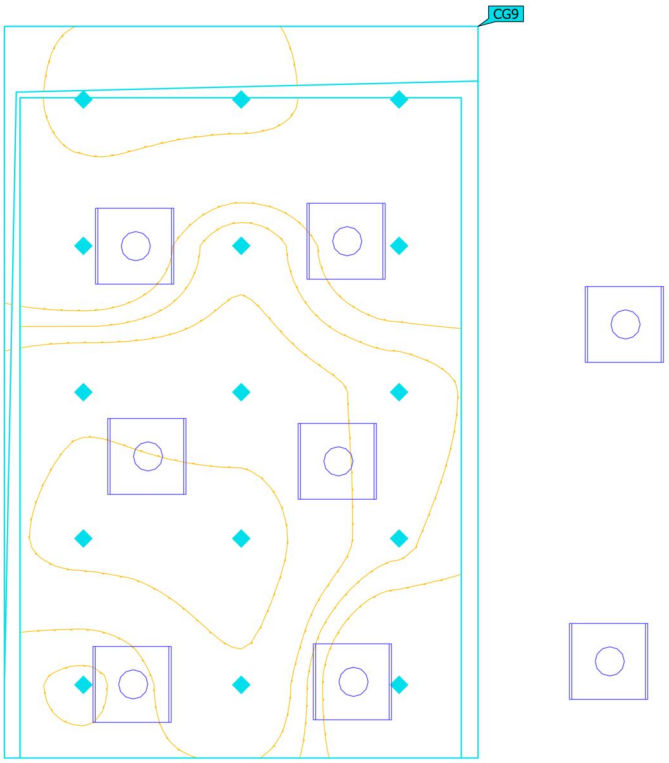
Properties	Ē	E_{min}	E_{max}	U_o (g_1)	g_2	Index
RECEPTION Perpendicular illuminance Height: 0.480 m	268 lx	4.54 lx	1462 lx	0.017	0.003	CG8

Utilisation profile: DIALux presetting (5.1.4 Standard (outdoor transportation area))

Site 1 (Light scene 1)

STAFFROOM



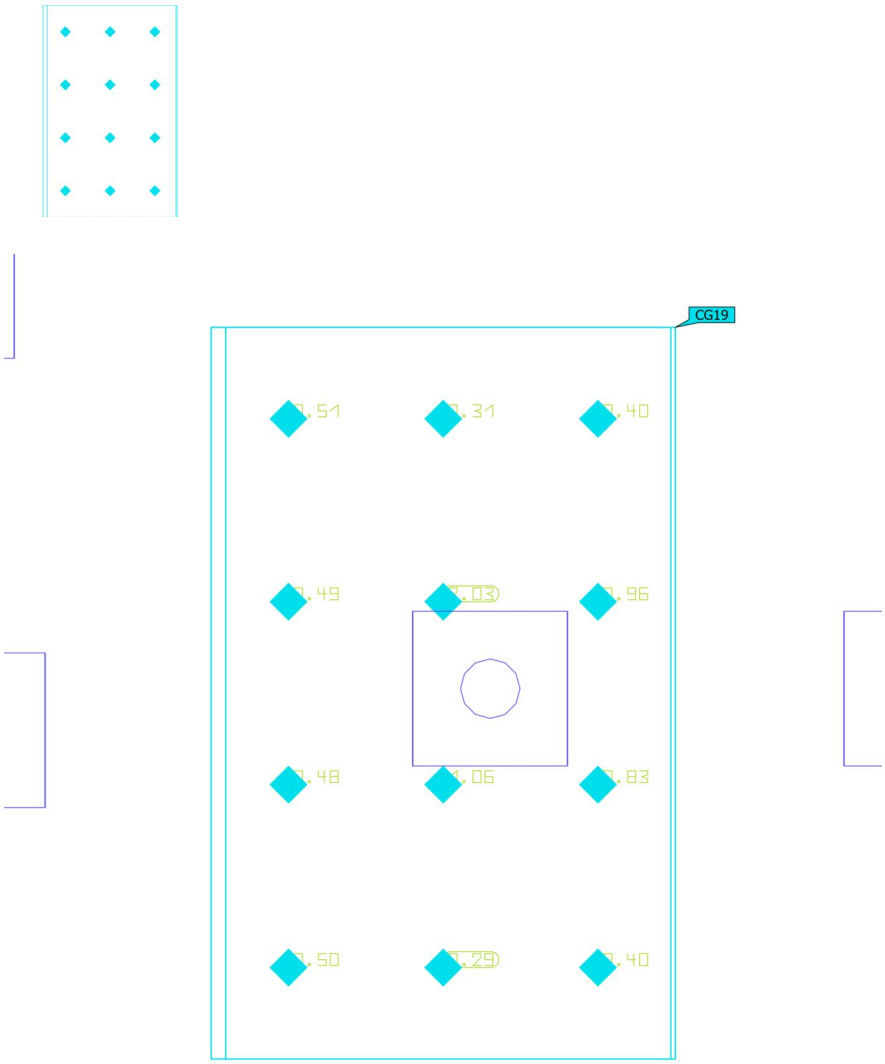


Properties	Ē	E_{min}	E_{max}	U_o (g_1)	g_2	Index
STAFFROOM Perpendicular illuminance Height: 0.635 m	7.79 lx	5.64 lx	10.4 lx	0.72	0.54	CG9

Utilisation profile: DIALux presetting (5.1.4 Standard (outdoor transportation area))

Site 1 (Light scene 1)

OFFICE 1



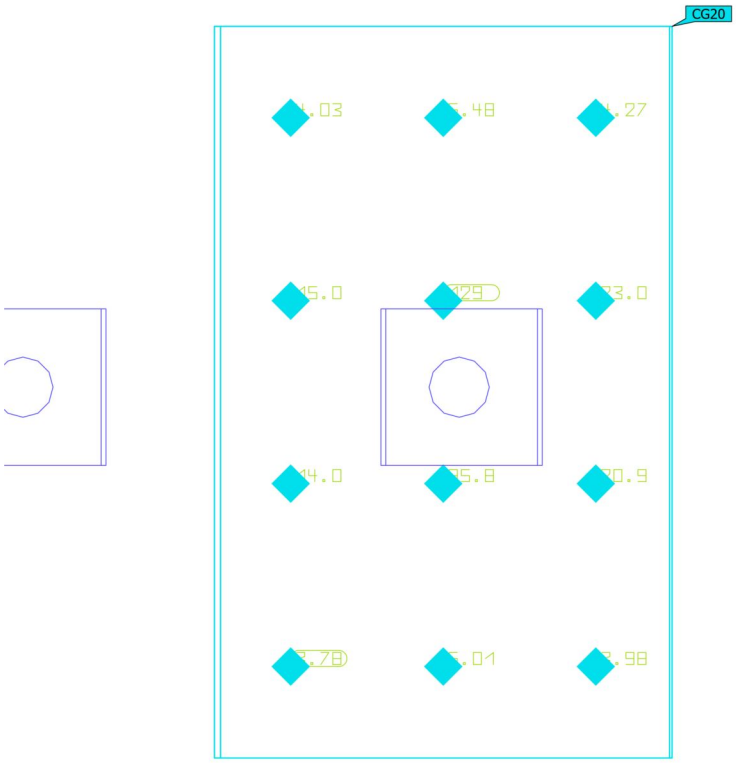
Properties	Ē	E_{min}	E_{max}	U_o (g_1)	g_2	Index
OFFICE 1 Perpendicular illuminance Height: 0.671 m	0.69 lx	0.29 lx	2.03 lx	0.42	0.14	CG19

Utilisation profile: DIALux presetting (5.1.4 Standard (outdoor transportation area))

Site 1 (Light scene 1)

OFFICE 2



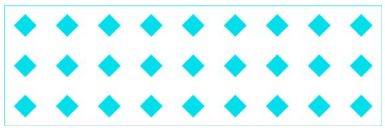


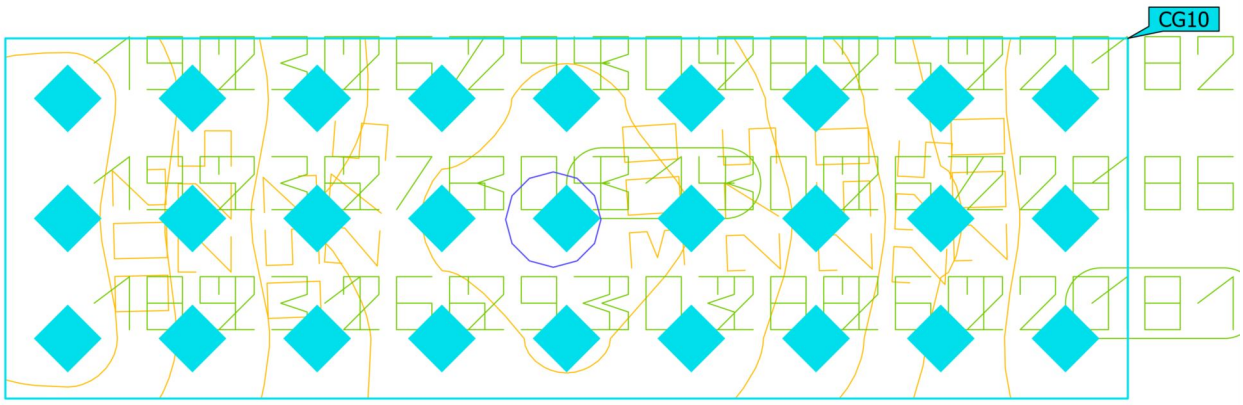
Properties	Ē	E_{min}	E_{max}	U_o (g_1)	g_2	Index
OFFICE 2 Perpendicular illuminance Height: 0.606 m	27.1 lx	3.78 lx	129 lx	0.14	0.029	CG20

Utilisation profile: DIALux presetting (5.1.4 Standard (outdoor transportation area))

Site 1 (Light scene 1)

SINK AREA



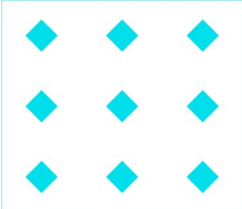


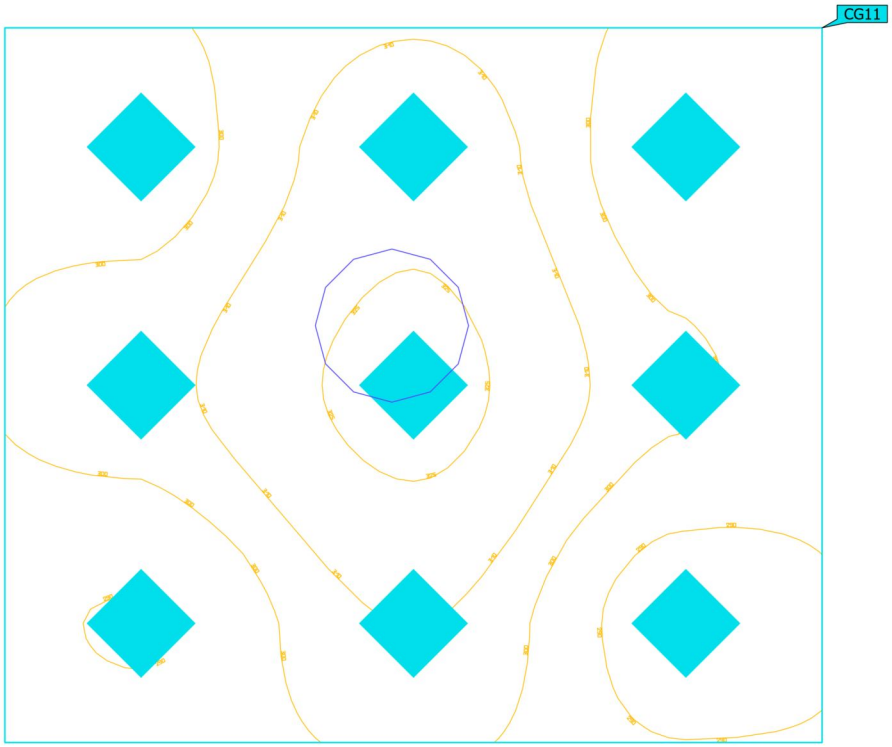
Properties	Ē	E_{min}	E_{max}	U_o (g_1)	g_2	Index
SINK AREA Perpendicular illuminance Height: 0.000 m	251 lx	181 lx	314 lx	0.72	0.58	CG10

Utilisation profile: DIALux presetting (5.1.4 Standard (outdoor transportation area))

Site 1 (Light scene 1)

URINAL AREA



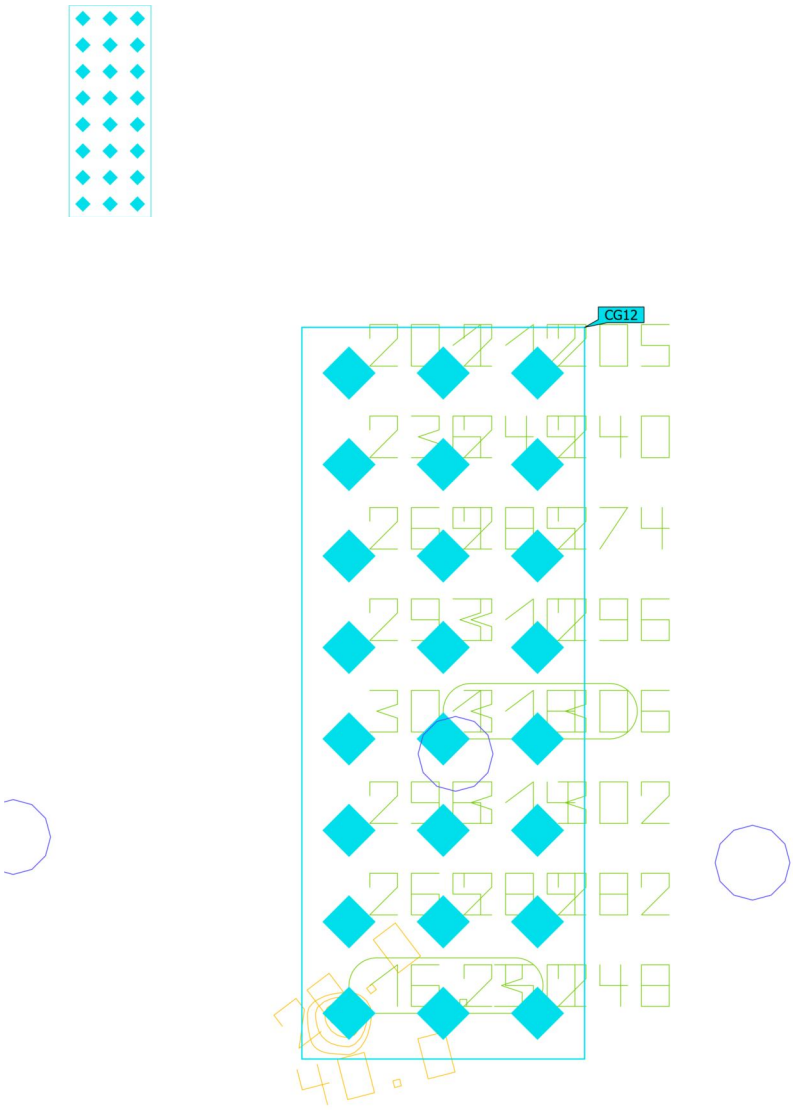


Properties	Ē	E_{min}	E_{max}	U_o (g_1)	g_2	Index
URINAL AREA Perpendicular illuminance Height: 0.000 m	303 lx	284 lx	331 lx	0.94	0.86	CG11

Utilisation profile: DIALux presetting (5.1.4 Standard (outdoor transportation area))

Site 1 (Light scene 1)

PWD WASHROOM

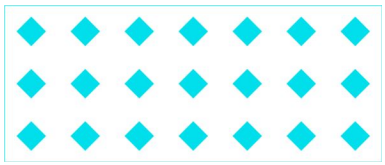


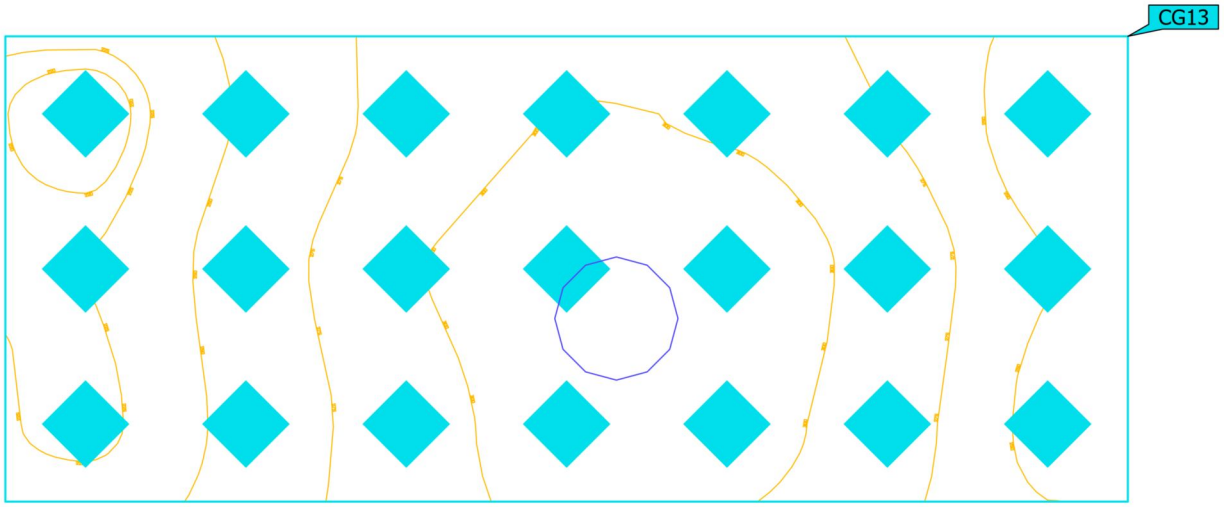
Properties	Ē	E_{min}	E_{max}	U_o (g_1)	g_2	Index
PWD WASHROOM Perpendicular illuminance Height: 0.000 m	261 lx	16.3 lx	318 lx	0.062	0.051	CG12

Utilisation profile: DIALux presetting (5.1.4 Standard (outdoor transportation area))

Site 1 (Light scene 1)

SINK AREA 2



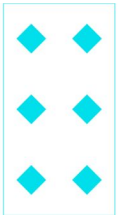


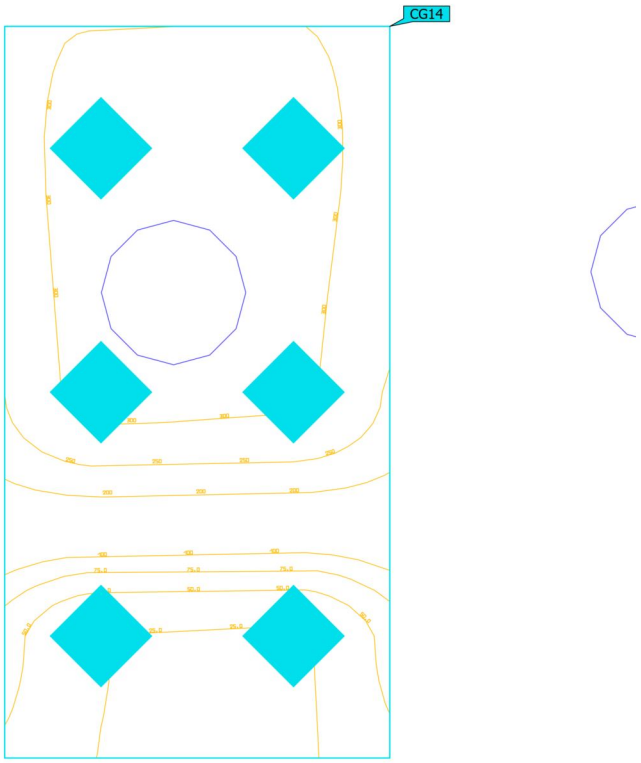
Properties	Ē	E_{min}	E_{max}	U_o (g_1)	g_2	Index
SINK AREA 2 Perpendicular illuminance Height: 0.000 m	274 lx	214 lx	318 lx	0.78	0.67	CG13

Utilisation profile: DIALux presetting (5.1.4 Standard (outdoor transportation area))

Site 1 (Light scene 1)

WASHROOM 1





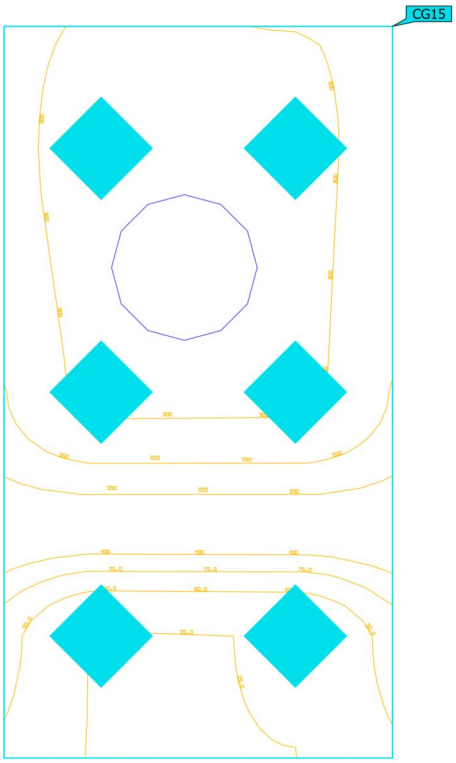
Properties	Ē	E_{min}	E_{max}	U_o (g_1)	g_2	Index
WASHROOM 1 Perpendicular illuminance Height: 0.000 m	214 lx	22.8 lx	314 lx	0.11	0.073	CG14

Utilisation profile: DIALux presetting (5.1.4 Standard (outdoor transportation area))

Site 1 (Light scene 1)

WASHROOM 2



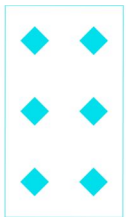


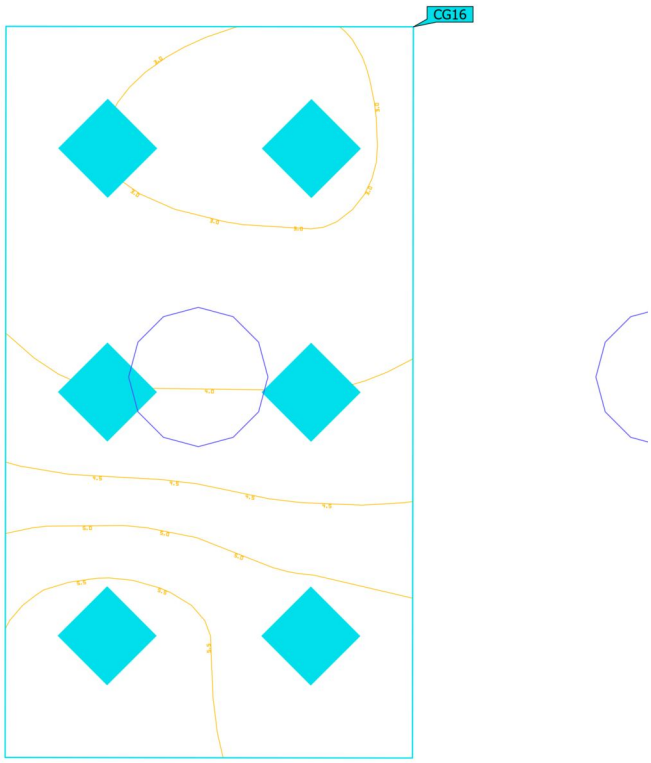
Properties	Ē	E_{min}	E_{max}	U_o (g_1)	g_2	Index
WASHROOM 2 Perpendicular illuminance Height: 0.000 m	214 lx	24.0 lx	310 lx	0.11	0.077	CG15

Utilisation profile: DIALux presetting (5.1.4 Standard (outdoor transportation area))

Site 1 (Light scene 1)

WASHROOM 3



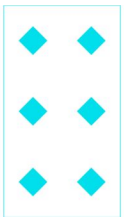


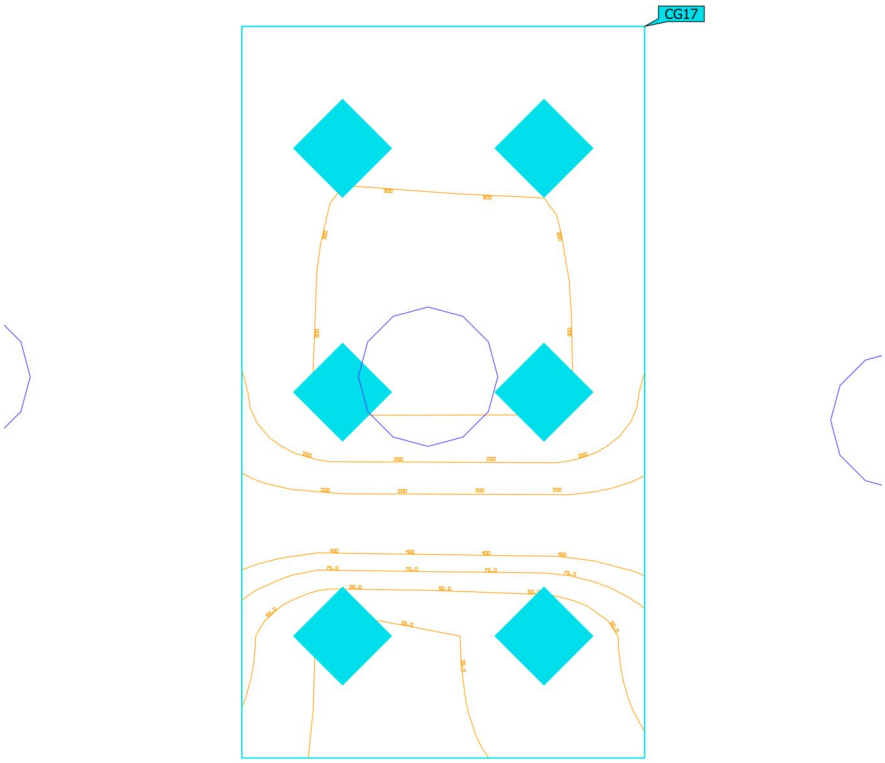
Properties	Ē	E_{min}	E_{max}	U_o (g_1)	g_2	Index
WASHROOM 3 Perpendicular illuminance Height: 0.500 m	4.13 lx	2.77 lx	5.75 lx	0.67	0.48	CG16

Utilisation profile: DIALux presetting (5.1.4 Standard (outdoor transportation area))

Site 1 (Light scene 1)

WASHROOM 4



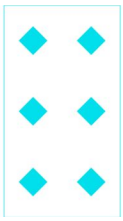


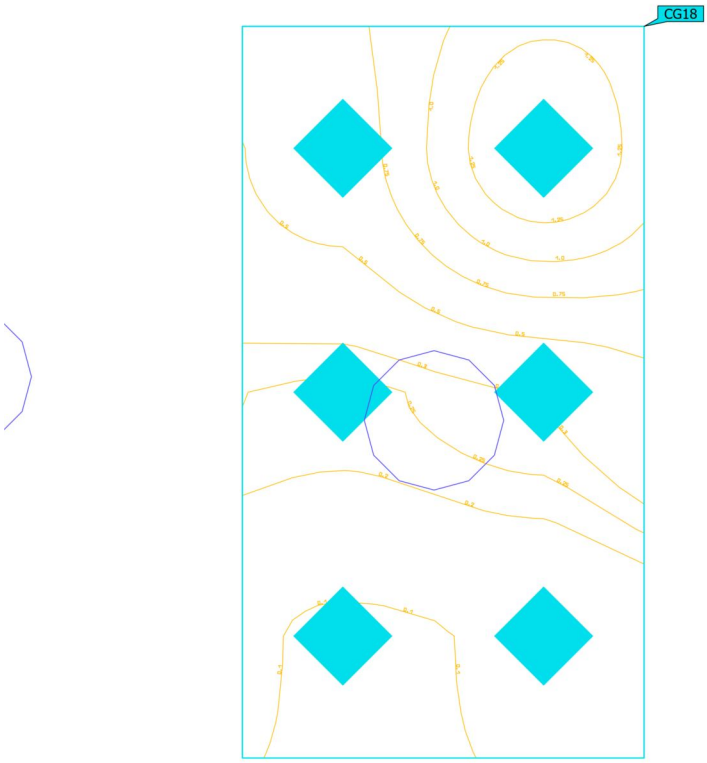
Properties	Ē	E_{min}	E_{max}	U_o (g_1)	g_2	Index
WASHROOM 4 Perpendicular illuminance Height: 0.000 m	210 lx	21.3 lx	308 lx	0.10	0.069	CG17

Utilisation profile: DIALux presetting (5.1.4 Standard (outdoor transportation area))

Site 1 (Light scene 1)

WASHROOM 5



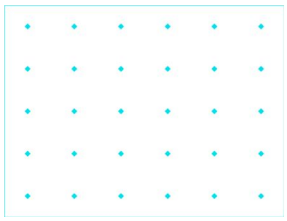


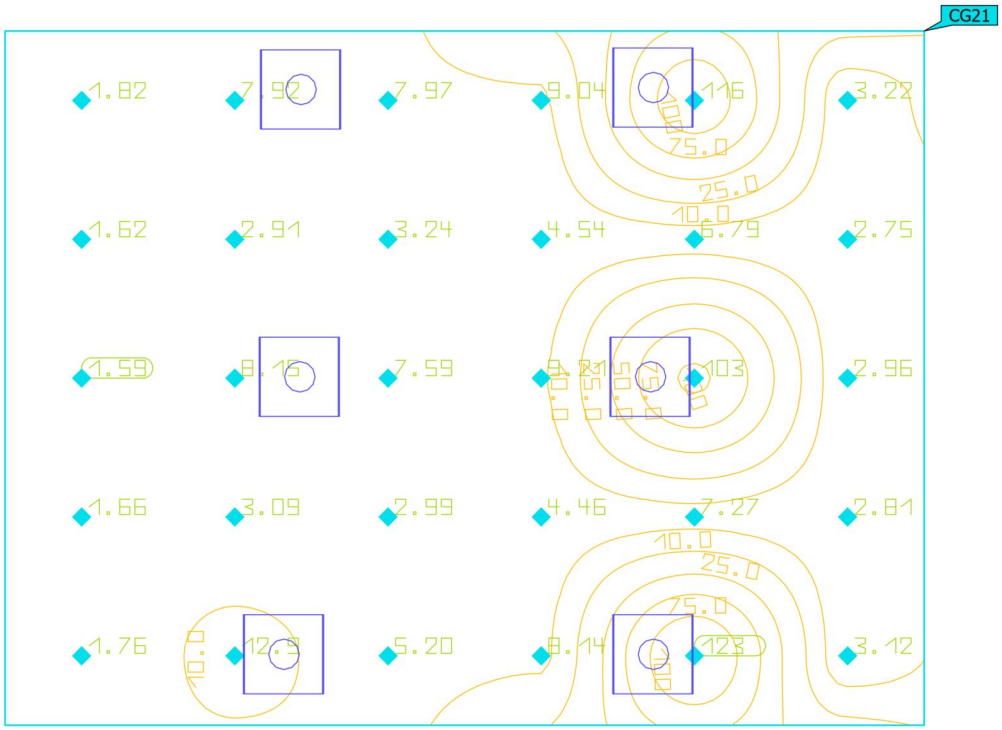
Properties	Ē	E_{min}	E_{max}	U_o (g_1)	g_2	Index
WASHROOM 5 Perpendicular illuminance Height: 0.500 m	0.47 lx	0.092 lx	1.50 lx	0.20	0.061	CG18

Utilisation profile: DIALux presetting (5.1.4 Standard (outdoor transportation area))

Site 1 (Light scene 1)

PRACTICAL ROOM 2





Properties	Ē	E_{min}	E_{max}	U_o (g_1)	g_2	Index
PRACTICAL ROOM 2 Perpendicular illuminance Height: 0.619 m	15.9 lx	1.59 lx	123 lx	0.10	0.013	CG21

Utilisation profile: DIALux presetting (5.1.4 Standard (outdoor transportation area))

Glossary

A

A	Formula symbol for a surface in the geometry

B

Background area	The background area borders the direct ambient area according to DIN EN 12464-1 and reaches up to the borders of the room. In larger rooms, the background area is at least 3 m wide. It is located horizontally at floor level.

C

CCT	(Engl. correlated color temperature) Body temperature of a thermal radiator which serves to describe its light color. Unit: Kelvin [K]. The lesser the numerical value the redder; the greater the numerical value the bluer the light colour. The color temperature of gas-discharge lamps and semi-conductors are termed "correlated color temperature" in contrast to the color temperature of thermal radiators. Allocation of the light colors to the color temperature ranges acc. to EN 12464-1: Light color - color temperature [K] warm white (ww) < 3,300 K neutral white (nw) ≥ 3,300 – 5,300 K daylight white (dw) > 5,300 K
Clearance height	The designation for the distance between upper edge of the floor and bottom edge of the ceiling (in the completely furnished status of room).
Control group	A group of luminaires that are dimmed and controlled together. For each lighting scene, a control group provides its own dimming value. All luminaires within a control group share this dimming value. The control groups with their luminaires are automatically determined by DIALux on the basis of the created light scenes and their luminaire groups.
CRI	(Engl. color rendering index) Designation for the color rendering index of a luminaire or a lamp acc. to DIN 6169: 1976 or CIE 13.3: 1995. The general color rendering index Ra (or CRI) is a dimensionless figure that describes the quality of a white light source in regards to its similarity with the remission spectra of defined 8 test colors (see DIN 6169 or CIE 1974) to a reference light source.

Glossary

D

Daylight autonomy	Describes what percentage of the daily working time the required illuminance is met by daylight. The nominal illuminance is used from the room profile, unlike described in EN 17037. The calculation is not done in the centre of the room but at the placed sensor measuring point. A room is considered sufficiently supplied with daylight if it achieves at least 50% daylight autonomy.
Daylight factor	Ratio of the illuminance achieved solely by daylight incidence at a point in the inside to the horizontal illuminance in the outer area under an unobstructed sky. Formula symbol: D (Engl. daylight factor) Unit: %
Daylight quotient effective area	A calculation surface within which the daylight quotient is calculated.

E

Energy evaluation	Based on an hourly calculation procedure for daylight in indoor spaces, considering the project geometry and any existing daylight control systems. Orientation and location of the project are also considered. The calculation uses the specified system power of the luminaires to determine the energy demand. A linear relationship between power and luminous flux in the dimmed state is assumed for daylight-controlled luminaires. Times of use and nominal illuminance are determined from the usage profiles of the spaces. Switched-on luminaires that are explicitly excluded from control also consider the specified times-of-use. The daylight control systems use a simplified control logic that closes them at an outdoor horizontal illuminance of 27,500lx. The calendar year 2022 is used as a reference only. It is not a simulation of this year. The reference year is only used to assign the days of the week to the calculated results. The changeover to summer time is not considered. The reference sky type used is the average sky described in CIE 110 without direct sunlight. The method was developed together with the Fraunhofer Institute for Building Physics and is available for review by the Joint Working Group 1 ISO TC 274 as an extension of the previous annual regression-based method.
Environmental zones	The assessment of intrusive light and light immission depends on the environment of the lighting installation. Depending on the standard, 4-6 different zones are defined, ranging from highly protected areas in natural settings to urban areas, commercial zones, and industrial zones.
Eta (η)	(light output ratio) The light output ratio describes what percentage of the luminous flux of a free radiating lamp (or LED module) is emitted by the luminaire when installed. Unit: %

Glossary

G

g_1	Often also U_o (Engl. overall uniformity) Designates the overall uniformity of the illuminance on a surface. It is the quotient from E_{min} to $\bar{E}$ and is required, for instance, in standards for illumination of workstations.
g_2	Actually it designates the "non-uniformity" of the illuminance on a surface. It is the quotient of E_{min} to E_{max} and is generally only relevant for certifying the emergency lighting acc. to EN 1838.

I

Illuminance	Describes the ratio of the luminous flux that strikes a certain surface to the size of this surface (lm/m^2 = lx). The illuminance is not tied to an object surface. It can be determined anywhere in space (inside or outside). The illuminance is not a product feature because it is a recipient value. Luxometers are used for measuring. Unit: Lux Abbreviation: lx Formula symbol: E
Illuminance, adaptive	For the determining of the middle adaptive illuminance on a surface, this is rastered "adaptively". In the area of large illuminance differences within the surface, the raster is subdivided finer; within lesser differences, a rougher classification is made.
Illuminance, horizontal	Illuminance that is calculated or measured on a horizontal (level) surface (this can be for example a table top or the floor). The horizontal illuminance is usually identified by the formula letter E_h.
Illuminance, perpendicular	Illuminance that is calculated or measured plumb-vertical to a surface. This needs to be taken into account for tilted surfaces. If the surface is horizontal or vertical, then there is no difference between the perpendicular and the horizontal or vertical illuminance.
Illuminance, vertical	Illuminance that is calculated or measured on a vertical surface (this can be for example the front of some shelves). The vertical illuminance is usually identified by the formula letter E_v.

K

k_S	The glare effect of a light source can be described by the glare metric k_S. It relates the solid angle of the glaring light source as seen from the point of immission, the ambient luminance, and the maximum allowable luminance.

Glossary

L

LENI	(Engl. lighting energy numeric indicator) Lighting energy numeric indicator acc. to EN 15193 Unit: kWh/(m^2 * a)
LLMF	(Engl. lamp lumen maintenance factor)/acc. to CIE 97: 2005 Lamp flux maintenance factor that takes the luminous flux reduction into account of a luminaire or an LED module in the course of the operating time. The lamp flux maintenance factor is specified as a decimal digit and can have a maximum value of 1 (no luminous flux reduction existing).
LMF	(Engl. luminaire maintenance factor)/acc. to CIE 97: 2005 Luminaire maintenance factor that takes the soiling into account of the luminaire in the course of the operating time. The luminaire maintenance factor is specified as a decimal digit and can have a maximum value of 1 (no soiling existing).
LSF	(Engl. lamp survival factor)/acc. to CIE 97: 2005 Lamp survival factor that takes the total failure into account of a luminaire in the course of the operating time. The lamp survival factor is specified as a decimal digit and can have a maximum value of 1 (no failures existing within the time concerned or prompt replacement after the failure).
Luminance	Dimension for the "brightness impression" that the human eye has of a surface. The surface itself can emit light thereby or light striking it can be reflected (emitter value). It is the only photometric value that the human eye can perceive. Unit: Candela per square metre Abbreviation: cd/m^2 Formula symbol: L
Luminous efficacy	Ratio of the emitted luminous flux Φ [lm] to the absorbed electrical power P [W] Unit: lm/W. This ratio can be formed for the lamp or LED module (lamp or module light output), the lamp or module with control gear (system light output) and the complete luminaire (luminaire light output).
Luminous flux	Dimension for the total light output that is emitted from one light source in all directions. It is thus an "emitter value" that specifies the entire emitting output. The luminous flux of a light source can only be determined in a laboratory. A difference is made between the lamp or LED module luminous flux and the luminaire luminous flux. Unit: Lumen Abbreviation: lm Formula symbol: Φ

Glossary

Luminous intensity	Describes the intensity of the light in a certain direction (emitter value). The luminous intensity is a matter of the luminous flux Φ that is emitted in a certain spherical angle Ω. The radiation characteristics of a light source are presented graphically in a light distribution curve (LDC). The luminous intensity is an SI base unit. Unit: Candela Abbreviation: cd Formula symbol: I

M

Maintenance factor	See MF
MF	(Engl. maintenance factor)/acc. to CIE 97: 2005 Maintenance factor as decimal number between 0 and 1 that describes the ratio of the new value of a photometric planning parameter (e.g. of the illuminance) to a maintenance value after a certain time. The maintenance factor takes into account the soiling of luminaires and rooms as well as the luminous flux reduction and the failure of light sources. The maintenance factor is taken into account either overall or determined in detail acc. to CIE 97: 2005 by the formula RMF x LMF x LLMF x LSF.

O

Obtrusive light/Light immission	To protect the nocturnal environment and minimize problems for humans, flora, and fauna, it is necessary to limit obtrusive light (also known as light pollution), which can cause serious physiological and ecological issues for individuals and the environment. Light immission refers to the disturbing influence of emitted light from artificial light sources.
Operating times	The assessment of obtrusive light and light immission depends on the operating times of the lighting installation. Depending on the standard, 1-3 different operating times are specified. In the absence of specific details, an operating time between 06:00 and 22:00 can be assumed.

P

P	(Engl. power) Electric power consumption Unit: watt Abbreviation: W

Glossary

R

$R_{(UG)}$ max	Measure of the psychological glare in indoor spaces. In addition to the luminance of luminaires, the level of the $R_{(UG)}$ value also depends on the observer position, the viewing direction and the ambient luminance. The calculation is made according to the table method, see CIE 117. Among other things, EN 12464-1:2021 specifies maximum permissible $R_{(UG)}$- values $R_{(UGL)}$ for various indoor workplaces.
R_{DLO}	The ratio of the luminous flux emitted below the horizontal plane to the total lamp luminous flux of a luminaire or lighting installation in its operational position.
R_G	The glare directly caused by luminaires of an outdoor lighting installation is determined using the CIE Glare Rating (RG) method. To calculate this, the equivalent veiling luminance of the surroundings is needed. There are four options for determining this: • An exact calculation according to CIE 112, based on the scene area. • A simplified method according to EN 12464-2, based on the scene area. • Using a custom calculation area to determine the equivalent veiling luminance. • Specifying a fixed value for easy comparability.
R_{UF}	upward flux ratio The ratio of the luminous flux emitted directly or reflected above the horizontal plane to the luminous flux that cannot be avoided under ideal conditions to achieve the illuminance level on a deliberately illuminated area.
R_{UL}	upward light ratio The ratio of the luminous flux emitted above the horizontal plane to the luminous flux of a luminaire or lighting installation in its operational position. The luminaire efficiency is considered in this calculation.
R_{ULO}	upward light output ratio The ratio of the luminous flux emitted above the horizontal plane to the total lamp luminous flux of a luminaire or lighting installation in its operational position.
Reflection factor	The reflection factor of a surface describes how much of the striking light is reflected back. The reflection factor is defined by the color of the surface.
RMF	(Engl. room maintenance factor)/acc. to CIE 97: 2005 Room maintenance factor that takes the soiling into account of the space encompassing surfaces in the course of the operating time. The room maintenance factor is specified as a decimal digit and can have a maximum value of 1 (no soiling existing).
RUG (max)	(unified glare rating) Measure for the psychological glare effect in interiors. In addition to luminaire luminance, the RUG value also depends on the position of the observer, the viewing direction and the ambient luminance. Among other things, EN 12464-1 specifies maximum permissible RUG values for various indoor workplaces.

Glossary

RUG observer	Calculation point in the room, for the DIALux the RUG value is determined. The location and height of the calculation point should correspond to the typical observer position (position and eye level of the user).

S

Surrounding area	The ambient area directly borders the area of the visual task and should be planned with a width of at least 0.5 m according to DIN EN 12464-1. It is at the same height as the area of the visual task.

V

Visual task area	The area that is needed for carrying out the visual task in accordance with DIN EN 12464-1. The height corresponds with the height at which the visual task is executed.

W

Wall zone	Circumferential area between working plane and walls which is not taken into account for the calculation.
Working plane	Virtual measuring or calculation surface at the height of the visual task that generally follows the room geometry. The working plane may also feature a wall zone.